General Relativity: Local Flatness, Covariant Differentiation, and Curvature

Based on lectures by Leonard Susskind

Lecture notes reconstructed and organized by LazyingArt LLC with Video2Book

Based on Leonard Susskind's Stanford lecture. The exposition and board record were reconstructed and organized by LazyingArt LLC with Video2Book.

Chapter 1

Local Flatness, Covariant Differentiation, and Curvature

Overview. A metric can always be made Euclidean through first order at one chosen point. Curvature begins where that coordinate freedom ends: in the second-order failure of neighboring local frames to fit together. We develop that statement in three stages. First come normal coordinates, then the covariant derivative and its connection, and finally the noncommutativity of two covariant derivatives, which defines the Riemann tensor and gives tidal gravity its invariant meaning.

1.1 The Necessary Burden of Indices

General relativity is difficult in a rather particular way. Its basic ideas are not impossibly obscure, but a calculation quickly becomes crowded with indices, derivatives, and symbols. More compressed languages exist: vielbeins, differential forms, spinors, twistors, and related devices can package long component equations into compact objects. Learning those languages, however, is itself a substantial task, and compact notation can conceal the machinery one most needs to understand on a first encounter.

Maxwell's equations offer the model. Written component by component, they amount to several separate equations. Vector notation collects them into divergence and curl equations, making the structure easier to see. General relativity admits analogous compressions, but the component method remains the dependable route when an explicit calculation must actually be done. We shall therefore keep the indices visible. Once their grammar becomes familiar, the conceptual path is much smoother, even though the arithmetic can still fill pages.

Symbolic computer packages can help with that arithmetic. Given a metric $g_{mn}(x)$, such a package can construct Christoffel symbols, Riemann tensors, Ricci tensors, and Einstein tensors. It can then substitute these expressions into field equations. This does not remove the need to understand what the objects mean. It merely separates the mechanical expansion from the geometry.

Question from the audience. What were the active areas of research in general relativity at the time of this lecture?

Response. The lecture singled out several fronts. Quantum gravity and the quantum physics of black holes were central problems in fundamental theory. Classical astrophysics and numerical relativity studied systems such as orbiting and merging black holes, where solving Einstein's equations and predicting gravitational radiation require much more than blindly feeding equations to a computer. The same mathematical structures were also finding uses in condensed-matter physics and fluid dynamics. The historical point was equally striking: general relativity had moved from a comparatively small specialty in the 1960s to one of the central languages of theoretical physics, tightly connected with quantum field theory. [a]

[a]Lecture timestamp: 00:05:15.

This is the last lecture devoted to Riemannian geometry by itself. Beginning with the next lecture, the geometry will be interpreted as gravity. The bridge is already visible. Distinguishing a genuine gravitational field from an apparent field caused by accelerated or curvilinear coordinates is mathematically the same problem as distinguishing a curved metric from a flat metric written in awkward coordinates.

1.2 The Invariant Question

A two-dimensional geometry is flat when Euclidean geometry works intrinsically: straight lines, parallel lines, triangle relations, and distance measurements all have their Euclidean form. It is tempting to say that flatness means $g_{mn} = \delta_{mn}$, but that statement is coordinate dependent. Even an ordinary plane has a position-dependent metric in polar coordinates. The correct statement is that a flat region admits coordinates in which the metric is constant and Euclidean throughout that region.

Suppose instead that the metric is handed to us in some arbitrary coordinates. One bad test would be to search through coordinate systems one after another, transforming the metric each time and hoping eventually to find a Cartesian form. There are infinitely many possibilities, so this is not a practical criterion.

The right strategy is to construct a diagnostic from g_{mn} and its derivatives. We want an object R with two decisive properties:

$$R = 0 \quad \Longleftrightarrow \quad \text{the geometry is locally flat,}$$

and

$$R \neq 0 \text{ at a point} \quad \Longrightarrow \quad \text{the geometry is curved there.}$$

Most importantly, R must be a tensor. A tensor that vanishes in one coordinate system vanishes in every coordinate system, so no coordinate search is needed.

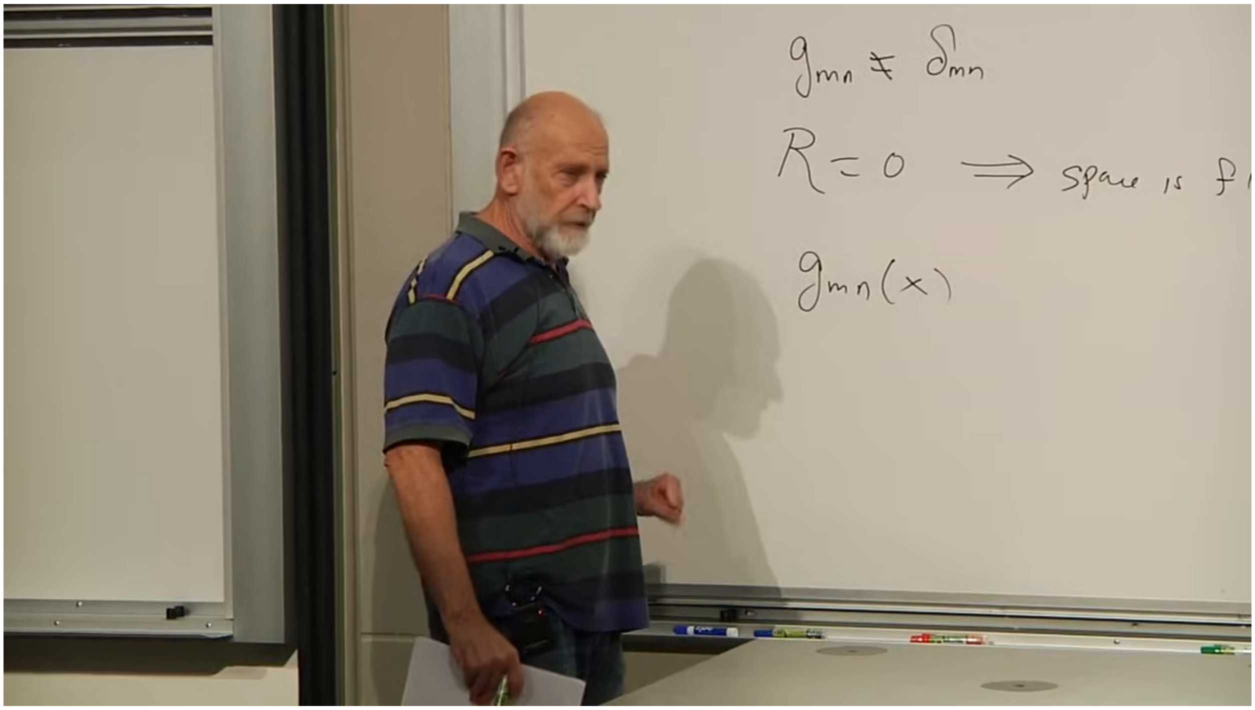

Figure 1.1: The desired invariant test: a metric need not visibly equal δ_{mn}, while a vanishing curvature object should identify a locally flat geometry.

Lecture frame: 00:18:20.

The qualification *locally* matters. A vanishing Riemann tensor guarantees local Euclidean geometry; global identifications can still produce nontrivial topology. A flat cylinder or flat torus, for example, can be locally Euclidean without being globally the plane.

1.3 Metric Geometry in the Small

It helps to see what changes with dimension. A zero-dimensional space is only a point. A connected one-dimensional Riemannian space can be straightened locally into a line; globally it may be an interval, a line, or a loop. A bug confined to a closed one-dimensional track can measure its circumference, but bending the track in an ambient room does not alter its intrinsic metric.

Two dimensions admit much richer intrinsic geometry: a plane, a sphere, a lumpy terrestrial surface, and a torus need not agree even locally, and their global topologies can differ as well. The metric is the instrument that records their infinitesimal distances:

$$ds^2 = g_{mn}(x)\, dx^m dx^n. \tag{1.1}$$

The quantity ds^2 is a scalar. Consequently the changes in dx^m under a coordinate transformation are exactly compensated by the transformation of g_{mn}.

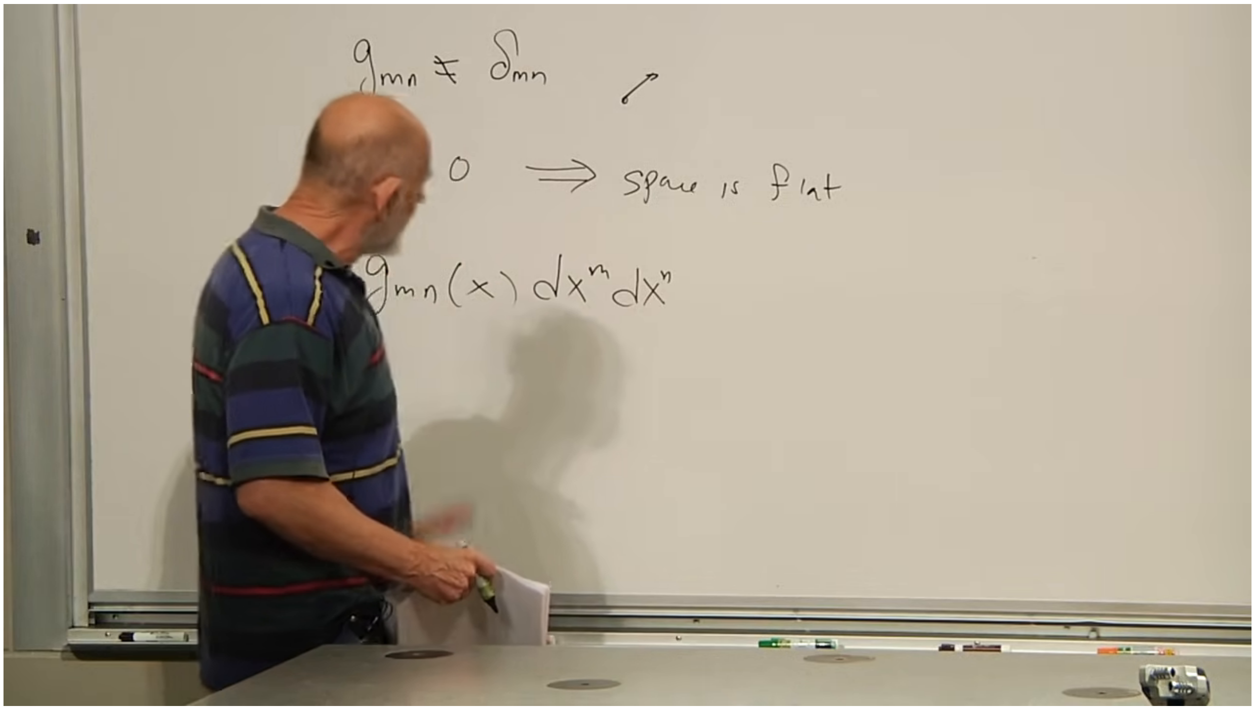

Figure 1.2: The invariant line element written with a position-dependent metric.
Lecture frame: 00:19:00.

The metric is nondegenerate and therefore has an inverse:

$$g^{mn} g_{nr} = \delta^m{}_r. \tag{1.2}$$

The mixed Kronecker tensor $\delta^m{}_r$ is the identity map. In ordinary Riemannian geometry the metric is also positive definite,

$$g_{mn} u^m u^n > 0 \qquad \text{for every nonzero } u^m. \tag{1.3}$$

Lorentzian spacetime will later relax this positivity while retaining nondegeneracy.

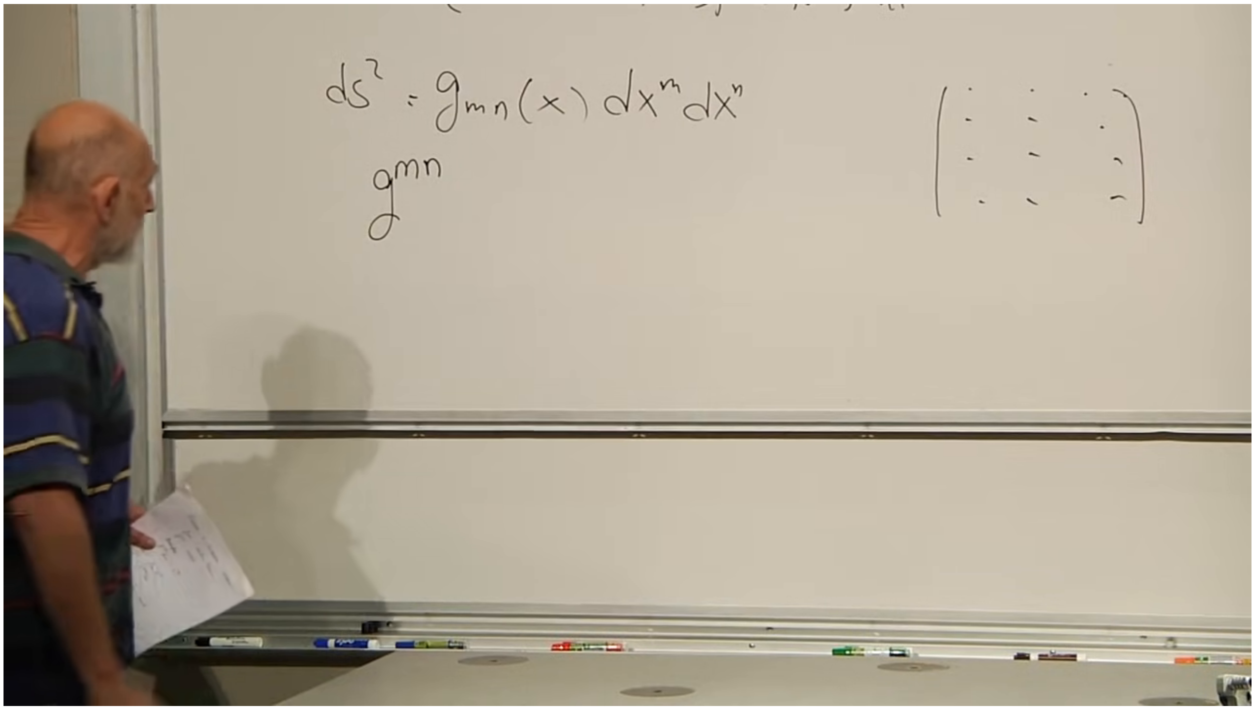

Figure 1.3: The inverse metric and the identity tensor on the board.
Lecture frame: 00:20:40.

The metric identifies vectors and covectors:

$$V_m = g_{mn} V^n, \qquad V^m = g^{mn} V_n. \tag{1.4}$$

Moving an index is therefore an operation performed by the metric, not a typographical change.

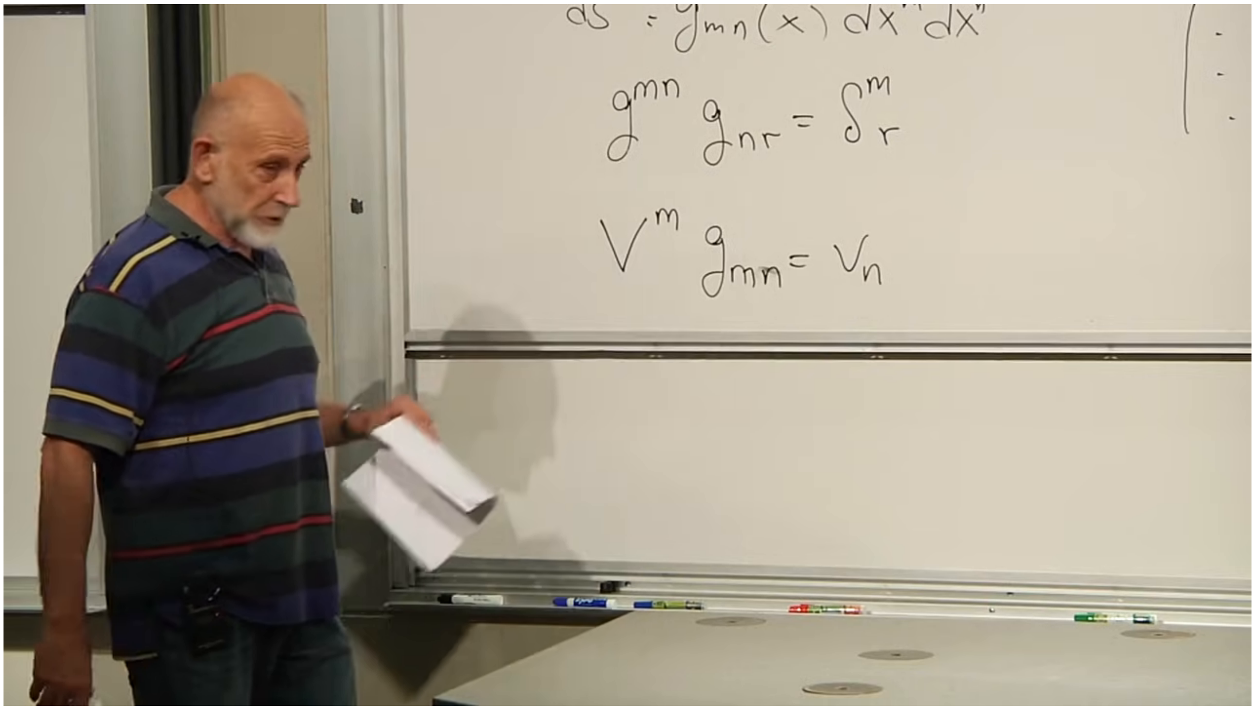

Figure 1.4: Raising and lowering a vector index with the metric and its inverse.
Lecture frame: 00:24:40.

If all components of g_{mn} are constant in a region, a constant linear change of coordinates can diagonalize and normalize the metric there. The geometry is then flat. The interesting information must therefore be tied to how the metric varies with position.

1.4 Normal Coordinates: Flat Through First Order

Choose a point P. The normal-coordinate theorem says that one can construct coordinates centered at P for which

$$g_{mn}(P) = \delta_{mn}, \tag{1.5}$$
$$\partial_r g_{mn}(P) = 0. \tag{1.6}$$

The lecture calls these Gaussian normal coordinates. Geometrically, one draws axes through P and extends their coordinate lines as straight as the geometry permits. The precise notion of straightness will become geodesic motion.

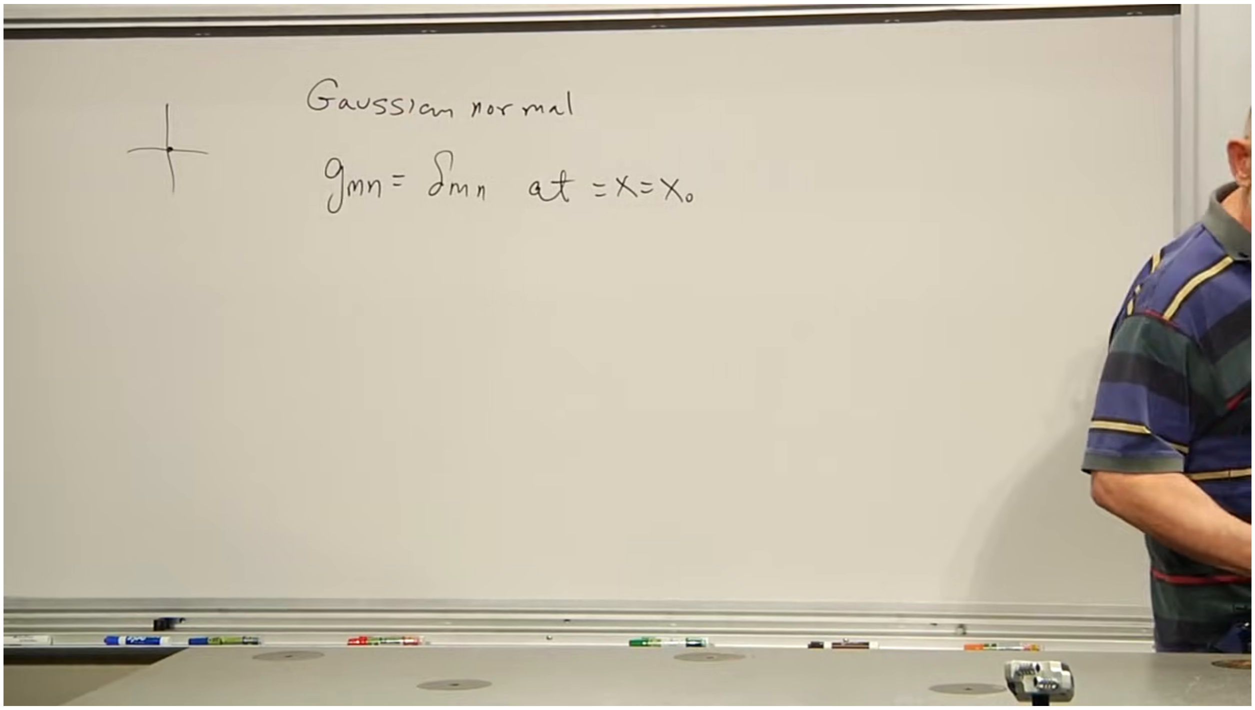

Figure 1.5: Normal coordinates chosen so that the metric is Euclidean at the selected point.
Lecture frame: 00:28:50.

There is no unique such frame. An orthogonal rotation at P preserves $g_{mn}(P) = \delta_{mn}$. What matters is not the orientation of the axes but the existence of a frame in which both (1.5) and (1.6) hold.

Question from the audience. Do these conditions hold only at the selected point, or throughout a neighborhood?

Response. They hold at the selected point P. One may center a normal frame at any point, but one generally cannot use a single coordinate system over a curved region and keep the metric Euclidean with all first derivatives zero everywhere. Extending the same coordinates away from P exposes higher derivatives of the metric. [a]

[a] Lecture timestamp: 00:29:53.

At P, the second derivatives are generally unavoidable:

$$\partial_r \partial_s g_{mn}(P) \neq 0 \qquad \text{in general.} \tag{1.7}$$

Thus the metric carries no invariant curvature information at zeroth or first order at one point; coordinate freedom can remove it. The first possible invariant information appears at second order.

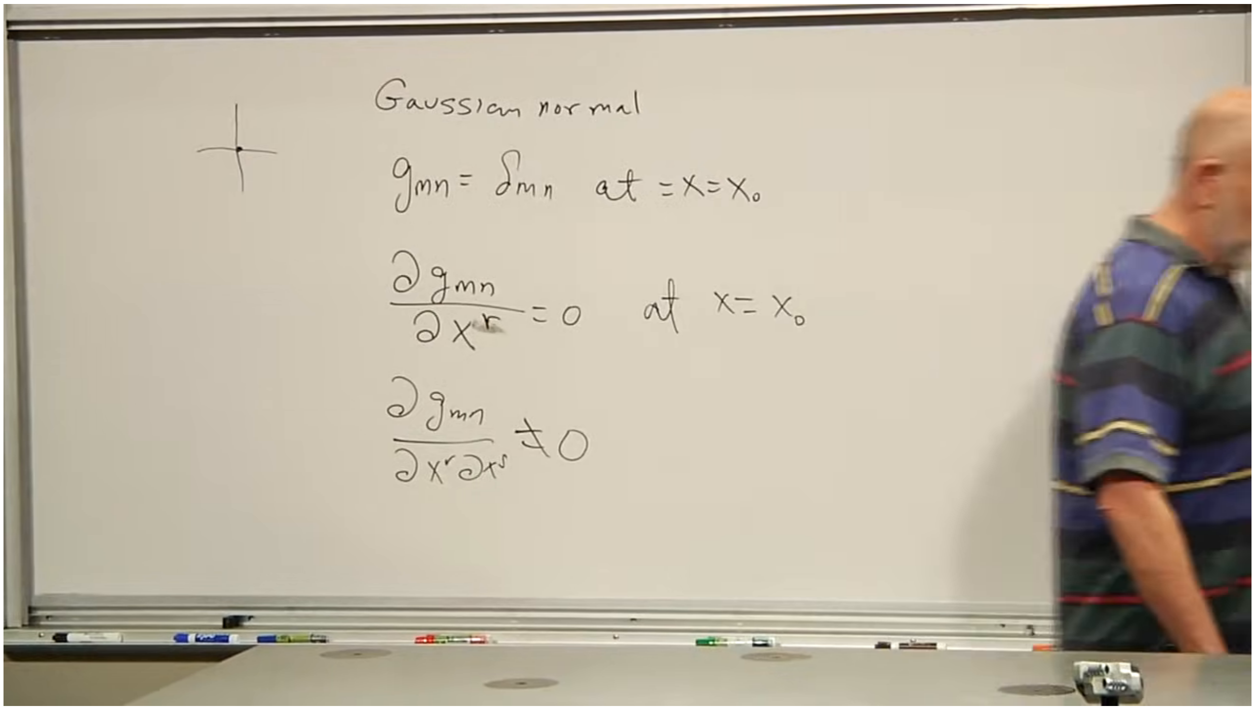

Figure 1.6: The normal-coordinate conditions: the metric is Euclidean, its first derivatives vanish, and its second derivatives need not.

Lecture frame: 00:30:50.

1.4.1 Why the coordinate freedom is sufficient

Take P to be the common origin of coordinates x^m and y^m. After using a linear transformation to arrange the metric at the origin, consider the quadratic part of a further coordinate change:

$$x^m = y^m + C^m{}_{nr} y^n y^r + O(y^3), \qquad C^m{}_{nr} = C^m{}_{rn}. \tag{1.8}$$

In four dimensions the symmetric pair nr has

$$\frac{4(4+1)}{2} = 10$$

independent choices. For each of the four values of m, there are ten coefficients $C^m{}_{nr}$, giving forty quadratic coefficients.

The symmetric metric also has ten independent components in four dimensions. Each component has four first derivatives, so the conditions $\partial_r g_{mn}(P) = 0$ comprise forty equations. The matching count shows that the quadratic coordinate freedom has exactly the right size to impose the first-derivative conditions. The normal-coordinate theorem supplies the existence statement; counting alone is a useful structural check rather than a complete proof of solvability for an arbitrary nonlinear system.

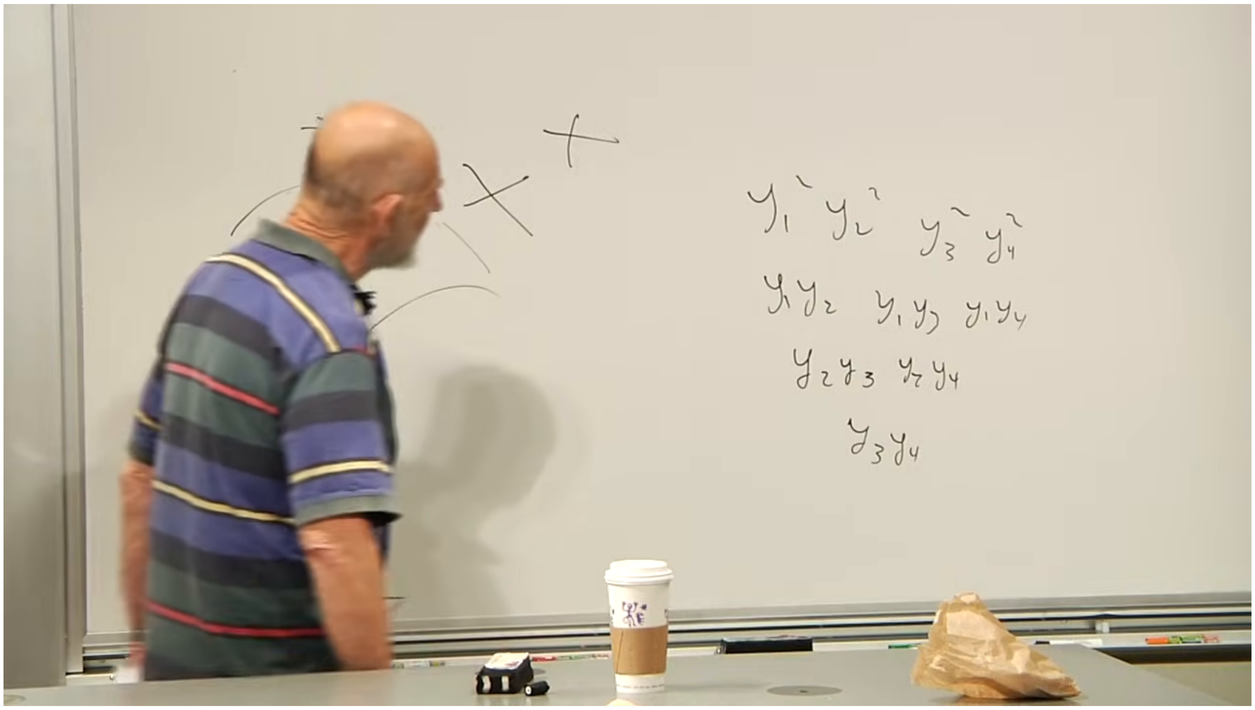

Figure 1.7: The quadratic coordinate transformation and the ten independent quadratic monomials in four dimensions.

Lecture frame: 00:33:50.

Question from the audience. Is $x^m = y^m + C^m{}_{nr} y^n y^r + \cdots$ an attempted coordinate transformation?

Response. Yes. The coefficients $C^m{}_{nr}$ are chosen so that, in the new coordinates, the first derivatives of the metric vanish at the common origin. The linear part keeps the Jacobian invertible near that point, while the quadratic part adjusts the first metric derivatives. [a]

[a]Lecture timestamp: 00:35:41.

Question from the audience. Is this what it means to say that a smooth geometry is locally flat?

Response. It is the relevant first-order statement. At a sufficiently small scale a smooth surface is approximated by its tangent plane, and normal coordinates express that fact through $g_{mn}(P) = \delta_{mn}$ and $\partial_r g_{mn}(P) = 0$. Exact flatness on a neighborhood is stronger: it also requires the curvature tensor to vanish there. [a]

[a]Lecture timestamp: 00:36:19.

Question from the audience. Does the equality between forty coefficients and forty equations really ensure that the equations can be solved?

Response. The count identifies the required coordinate freedom, while the theorem guarantees the local construction for a smooth nondegenerate metric. The lecture proposes checking the two-dimensional case directly as an exercise. The argument does not depend on four dimensions; in d dimensions the same symmetry and coordinate freedom fit together. [a]

[a]Lecture timestamp: 00:36:45.

Question from the audience. Why is there no constant term in the expansion, and which term is first order?

Response. The term y^m is the first-order term. A constant term was omitted because the two coordinate origins were deliberately chosen to represent the same point, so $y = 0$ implies $x = 0$. That choice costs no generality. [a]

[a]Lecture timestamp: 00:38:52.

1.5 Why Componentwise Differentiation Fails

Now consider a covariant vector field $V_m(x)$. In straight Cartesian coordinates on flat space, a geometrically constant vector has constant components. One might therefore try to define its derivative as $\partial_r V_m$. The difficulty appears as soon as the coordinate grid bends.

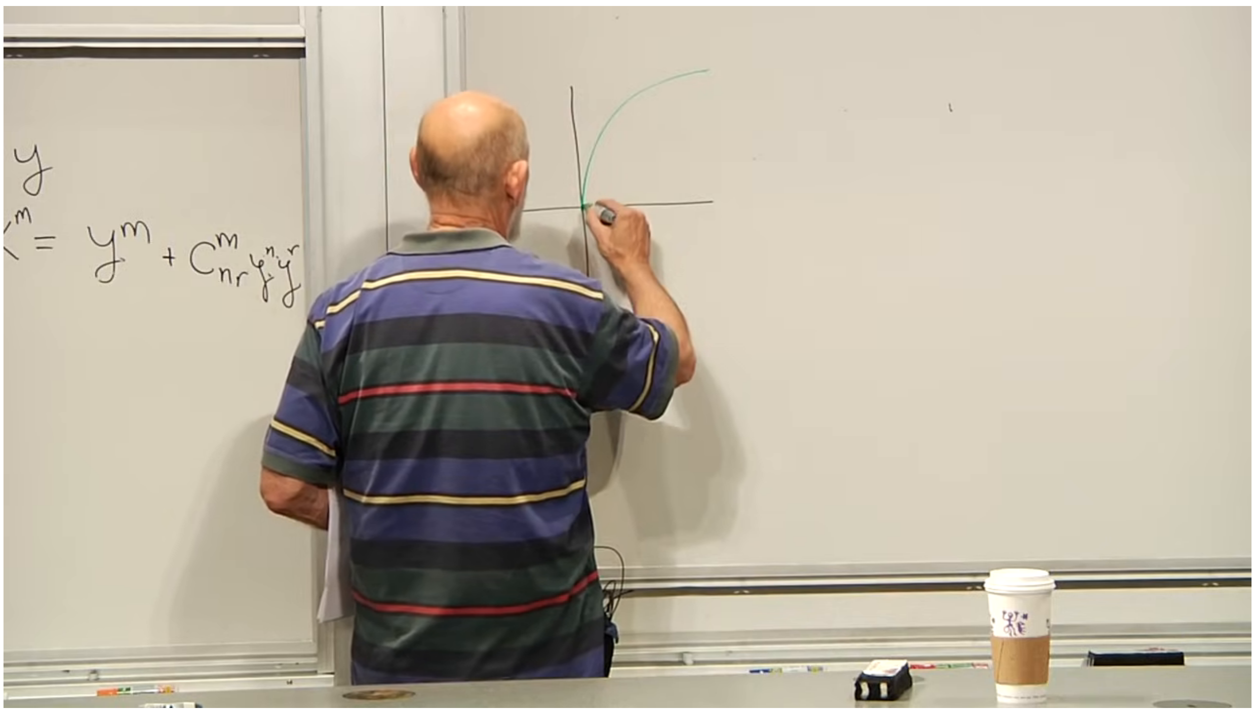

Figure 1.8: A curvilinear coordinate grid superimposed on a locally straight frame.

Lecture frame: 00:40:10.

Place the same geometric vector at neighboring points. Relative to a fixed Cartesian frame the components do not change. Relative to a curvilinear basis, however, the basis vectors themselves turn as one moves, so the component list changes even though the vector does not.

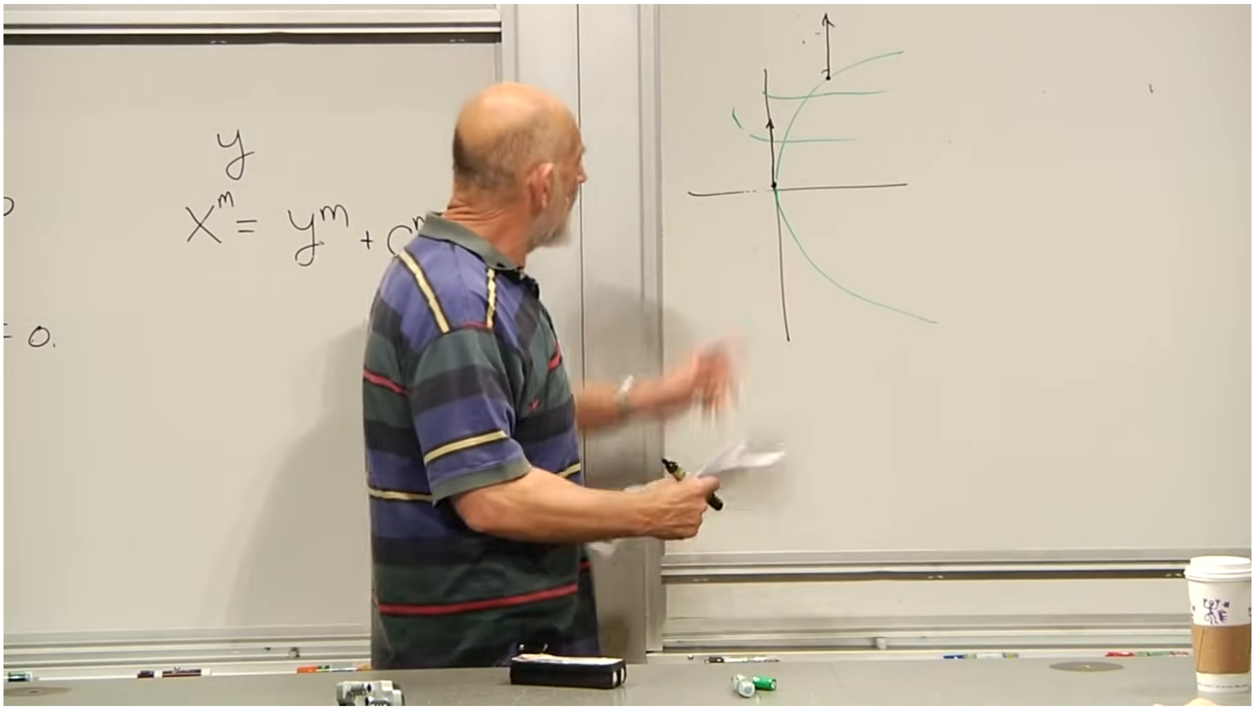

Figure 1.9: The same vector at neighboring points acquires different components because the local coordinate basis has rotated.

Lecture frame: 00:42:30.

The naive condition

$$\partial_r V_m = 0 \tag{1.9}$$

can therefore be true in one coordinate system and false in another. If $\partial_r V_m$ were a rank-two tensor, that would be impossible: vanishing in one frame would imply vanishing in every frame. Ordinary partial derivatives of vector components are consequently not tensor components.

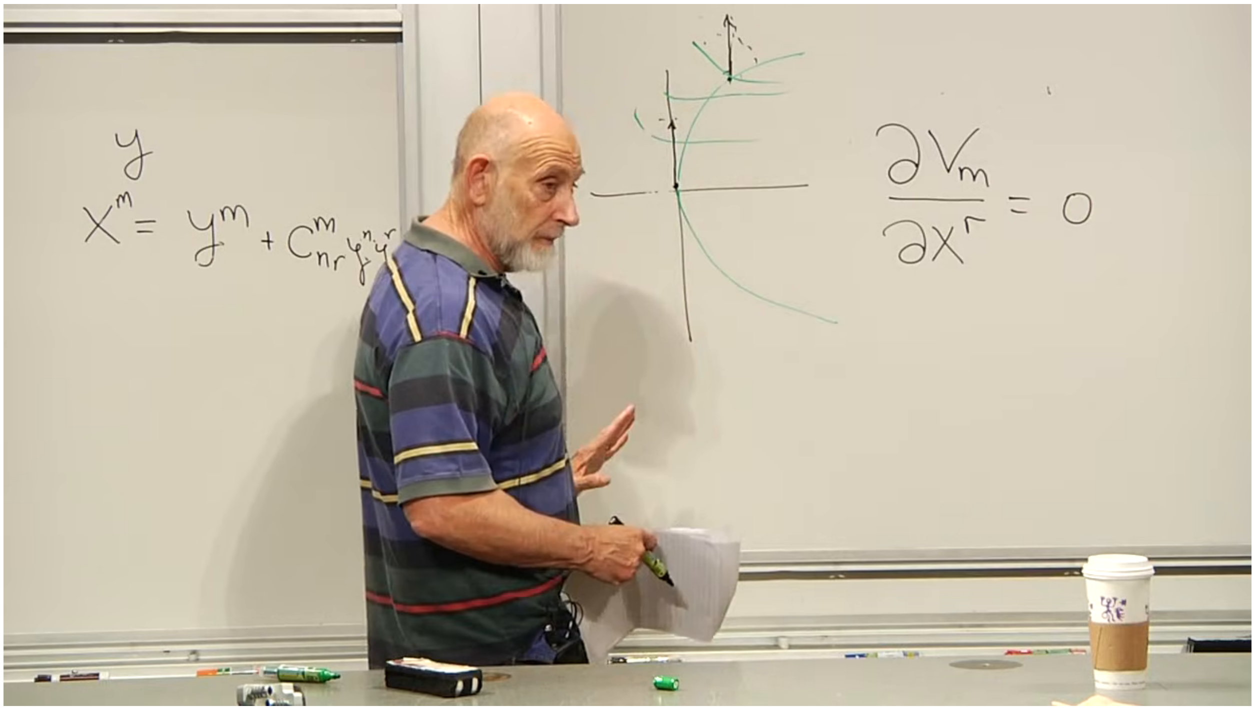

Figure 1.10: The deliberately naive zero-derivative condition beside the changing curvilinear basis.
Lecture frame: 00:44:00.

We need a derivative that removes the apparent change caused by the basis while retaining the genuine change of the vector field.

1.6 Defining the Covariant Derivative

Differentiation at P only compares values in an arbitrarily small neighborhood of P. We may therefore begin in normal coordinates centered there. Take a vector at a neighboring point, shift its tail back to P, and keep its normal-coordinate components fixed during the shift. Comparing the transported vector with the vector already at P, then dividing by the separation, gives the ordinary derivative in that locally straight frame.

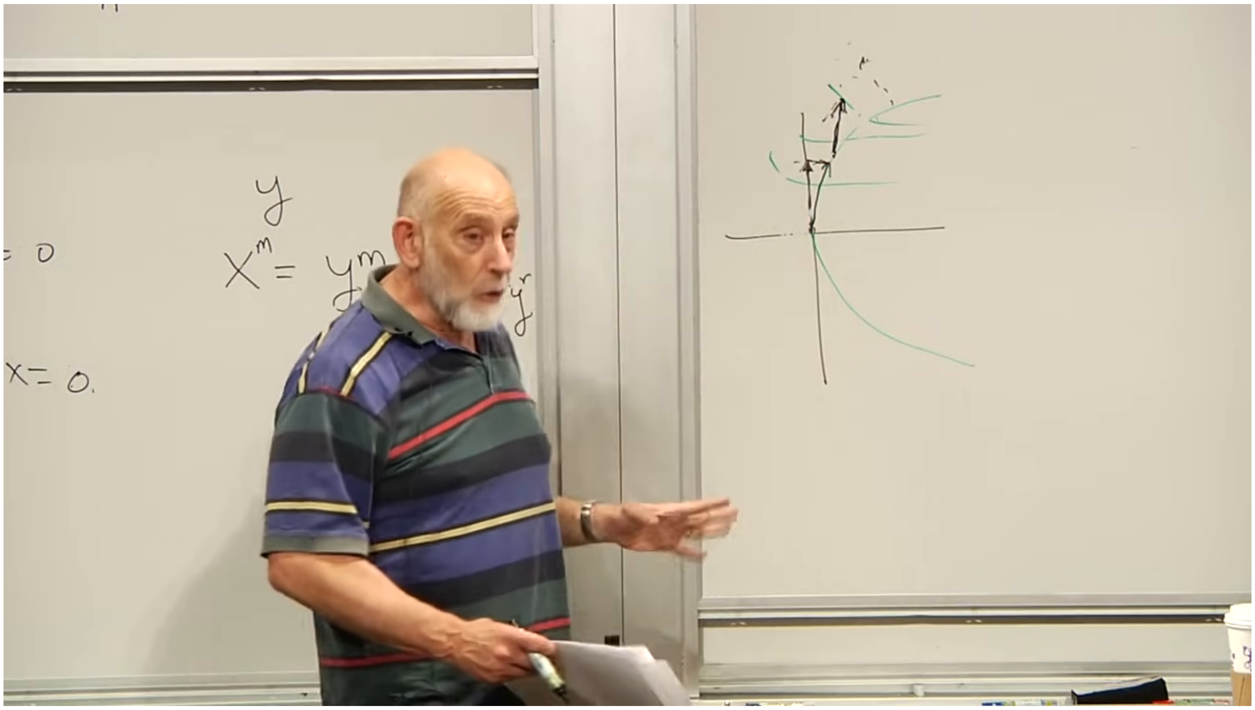

Figure 1.11: The local comparison: translate the neighboring vector in normal coordinates before taking the difference.

Lecture frame: 00:46:50.

Now express that geometrically defined derivative in arbitrary coordinates. There are two contributions: the components V_m may change, and the coordinate basis may change. Index structure fixes the form of the correction:

$$\nabla_r V_m = \partial_r V_m - \Gamma^t{}_{rm} V_t. \tag{1.10}$$

The quantities $\Gamma^t{}_{rm}$ are the connection coefficients, also called Christoffel symbols. The minus sign is appropriate for a lower vector index. At the center P of normal coordinates,

$$\Gamma^t{}_{rm}(P) = 0, \qquad \nabla_r V_m(P) = \partial_r V_m(P). \tag{1.11}$$

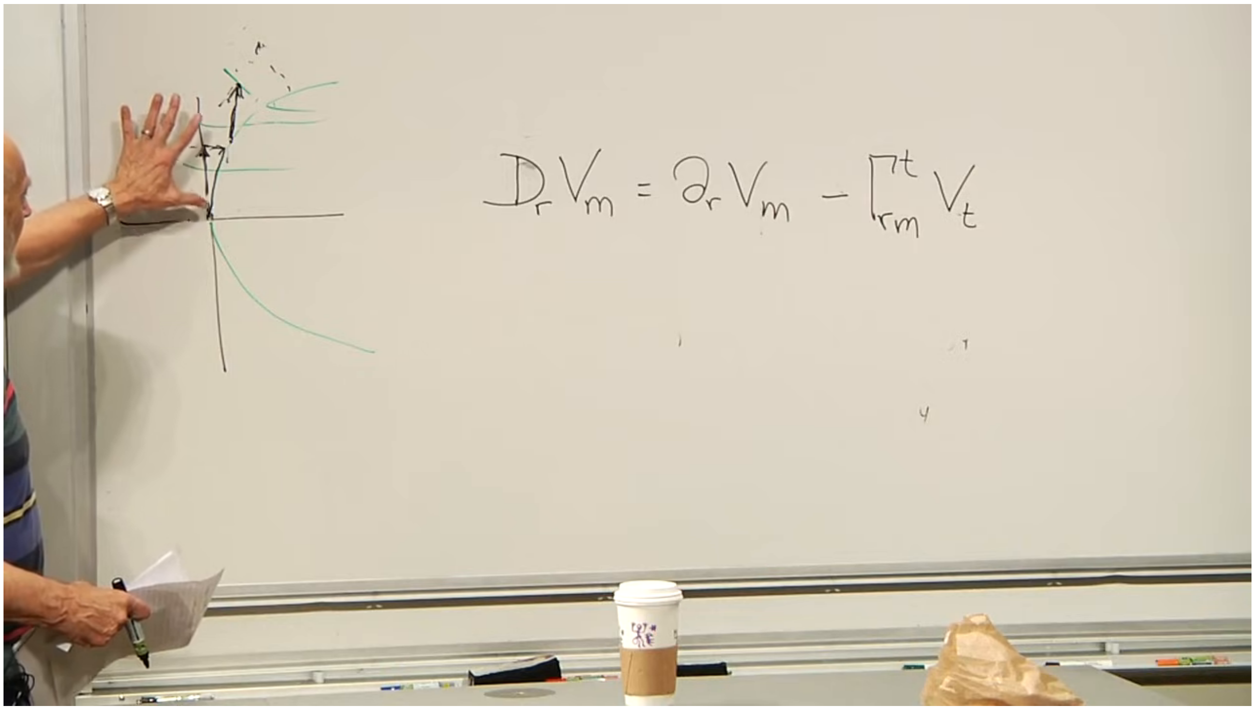

Figure 1.12: The covariant derivative of a covector: ordinary component change minus the change induced by the moving basis.

Lecture frame: 00:49:50.

Question from the audience. Does $\nabla_r V_m = \partial_r V_m - \Gamma^t{}_{rm} V_t$ hold in arbitrary coordinates, and what does it become in normal coordinates?

Response. The full formula holds in arbitrary coordinates. At the selected point in normal coordinates the first metric derivatives, and hence the Christoffel symbols, vanish, leaving only $\partial_r V_m$. Away from that point, even the same normal-coordinate chart generally has nonzero connection coefficients. [a]

[a] Lecture timestamp: 00:50:22.

The covariant derivative is so named because the entire object $\nabla_r V_m$ transforms covariantly as a tensor; the name is not merely a comment about the lower derivative index. The Christoffel term is precisely what repairs the nontensorial transformation of the partial derivative.

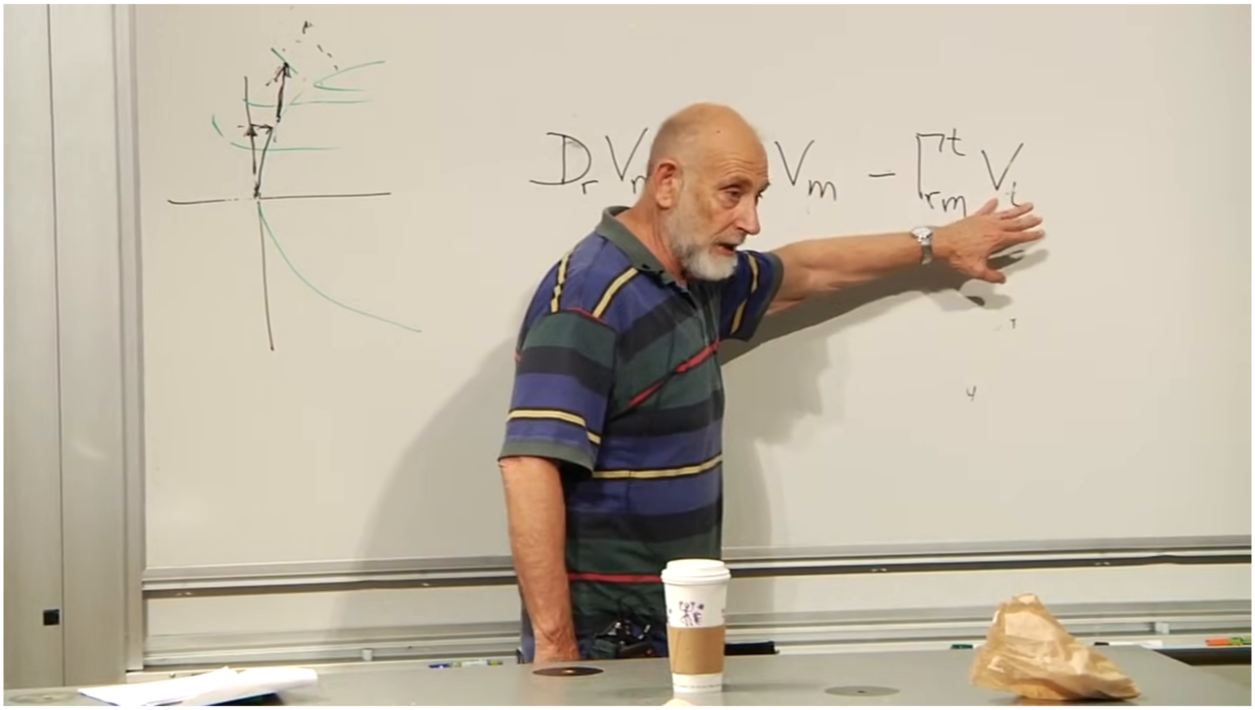

Figure 1.13: The complete covector derivative and its connection correction.
Lecture frame: 00:51:30.

Question from the audience. Is this related to the convective derivative used in fluid flow?

Response. The lecture answers cautiously that there is a connection: fluid mechanics also distinguishes change carried by a flow from change seen at a fixed coordinate point, and covariant derivatives occur in fluid equations on curved coordinates or curved spaces. The operators are not identical by definition, but both correct a naive derivative for structure that is moving with the description. [a]

[a]Lecture timestamp: 00:52:31.

For the torsion-free connection used here,

$$\Gamma^{t}{}_{rm} = \Gamma^{t}{}_{mr}. \tag{1.12}$$

More general geometries can possess torsion and need not obey this symmetry. The present connection is the one relevant to the standard metric formulation of general relativity.

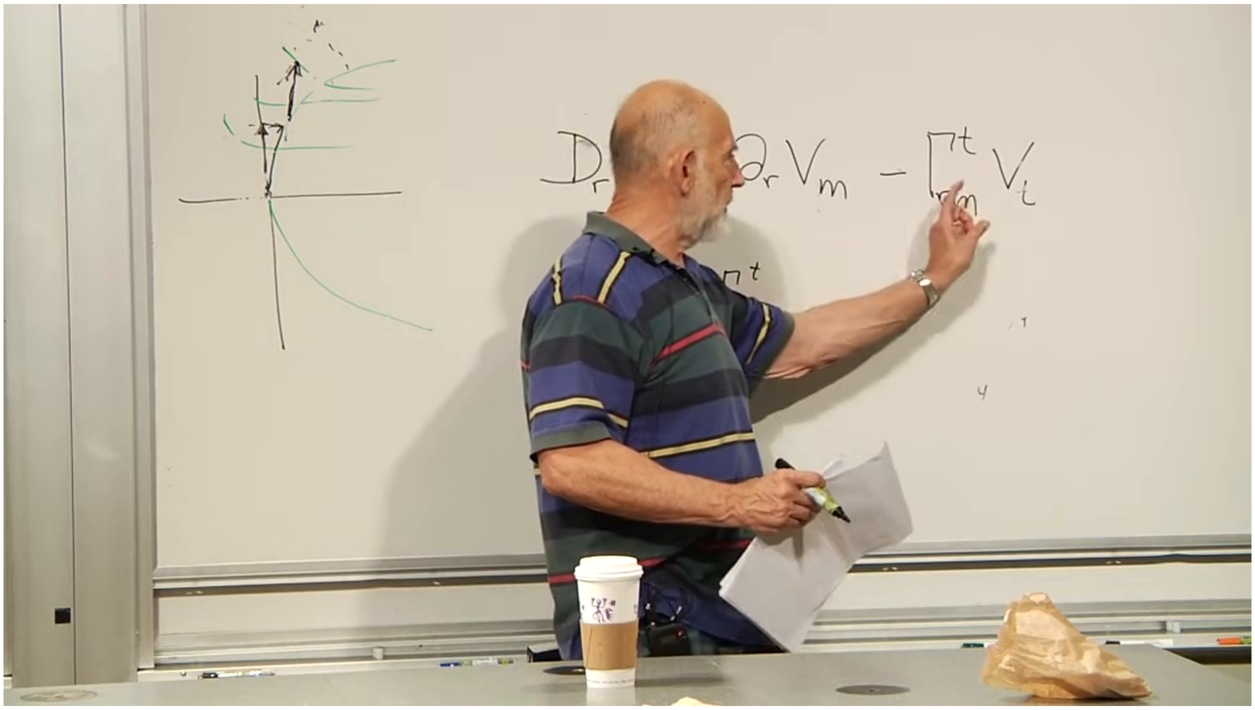

Figure 1.14: The Christoffel symbols identified as the symmetric connection coefficients of the torsion-free geometry.

Lecture frame: 00:54:30.

There is a useful physical analogy. Normal coordinates play the role of a freely falling frame at one event: the coordinate correction can be made to disappear there. Arbitrary curvilinear coordinates resemble an accelerated elevator, where apparent forces enter because the frame itself changes. The analogy is local, and curvature will measure why a single freely falling frame cannot remove gravity throughout an extended region.

1.6.1 One correction for each tensor index

For a covariant rank-two tensor T_{mn}, each lower index receives its own connection correction:

$$\nabla_s T_{mn} = \partial_s T_{mn} - \Gamma^t{}_{sm} T_{tn} - \Gamma^t{}_{sn} T_{mt}. \tag{1.13}$$

To remember the rule, hold one index fixed as a spectator while differentiating the other, and then repeat for the second index. Contravariant indices receive corresponding terms with the opposite sign.

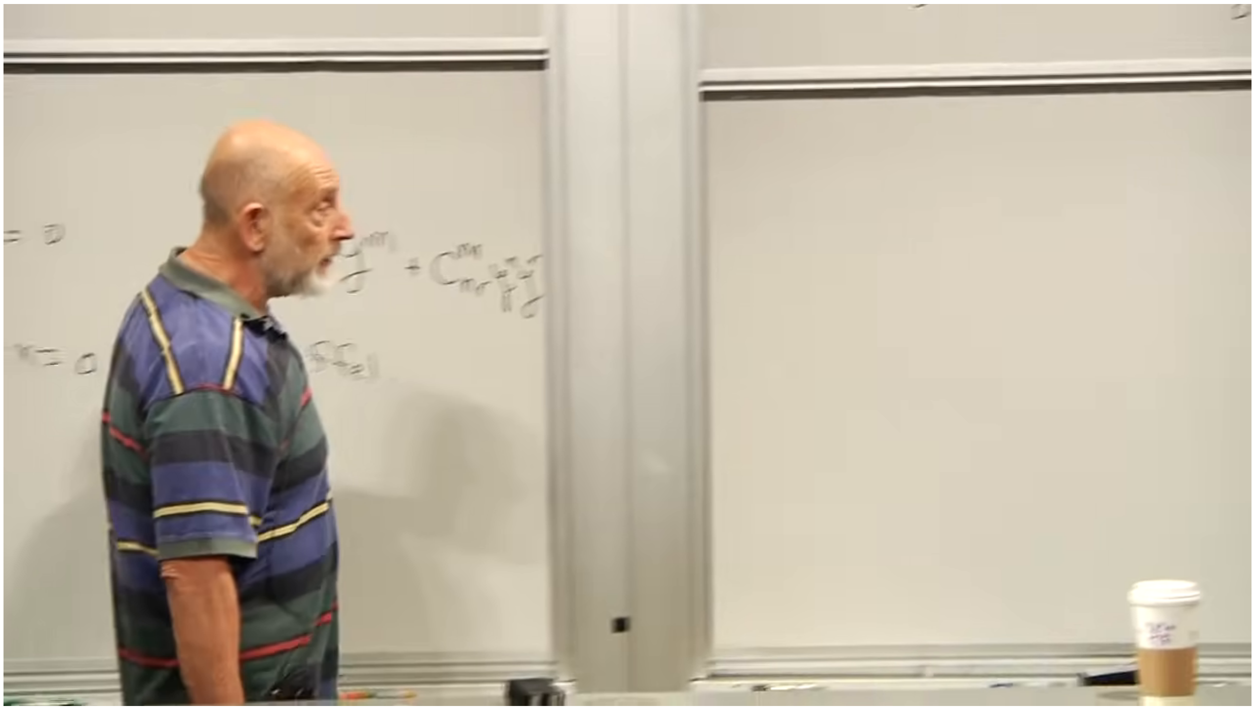

Figure 1.15: The covariant derivative of a rank-two covariant tensor, with one Christoffel correction for each lower index.

Lecture frame: 01:01:50.

Question from the audience. Does the symbol ∂_r itself refer only to normal coordinates, and is the tensor formula restricted to that frame?

Response. No. The notation $\partial_r = \partial/\partial x^r$ is meaningful in any coordinate system, and the full covariant-derivative formula is written for arbitrary coordinates. Only at the chosen point of a normal frame do the Christoffel corrections vanish. [a]

[a]Lecture timestamp: 00:57:22.

Question from the audience. When the covariant derivative reduces to the partial derivative in normal coordinates, is that reduction only at the selected point? Are the Christoffel symbols tensors?

Response. Yes, the reduction is pointwise. The Christoffel symbols are not tensors: they can vanish at P in normal coordinates and be nonzero at the same point in another coordinate system. Their nontensorial transformation is exactly what allows the combination in (1.10) to transform as a tensor. [a]

[a]Lecture timestamp: 00:58:13.

Question from the audience. What will the covariant derivative actually be used for?

Response. For field equations. Gravitational fields vary from point to point, so their equations must contain derivatives. At the same time, the equations should have the same content in every coordinate system. Covariant differentiation turns tensors into tensors and therefore permits differential equations whose truth does not depend on a peculiar reference frame. [a]

[a]Lecture timestamp: 01:02:35.

Question from the audience. Can the covariant derivative be multiplied by a coordinate differential to describe the change between neighboring points?

Response. Yes. Along an infinitesimal displacement dx^s, the covariant change is $dx^s \nabla_s T$. The partial term measures the change of components, while the connection terms account for the change of the coordinate basis. Even flat space in polar coordinates requires that distinction. [a]

[a]Lecture timestamp: 01:03:22.

1.7 Metric Compatibility Determines the Connection

Apply the rank-two rule to the metric. At P in normal coordinates, $\partial_s g_{mn} = 0$ and $\Gamma^t{}_{sm} = 0$, so $\nabla_s g_{mn} = 0$. Since the covariant derivative is a tensor, this pointwise result is true in every coordinate system:

$$\nabla_s g_{mn} = 0. \tag{1.14}$$

This is metric compatibility. It says that parallel transport preserves the metric, and hence preserves lengths and angles.

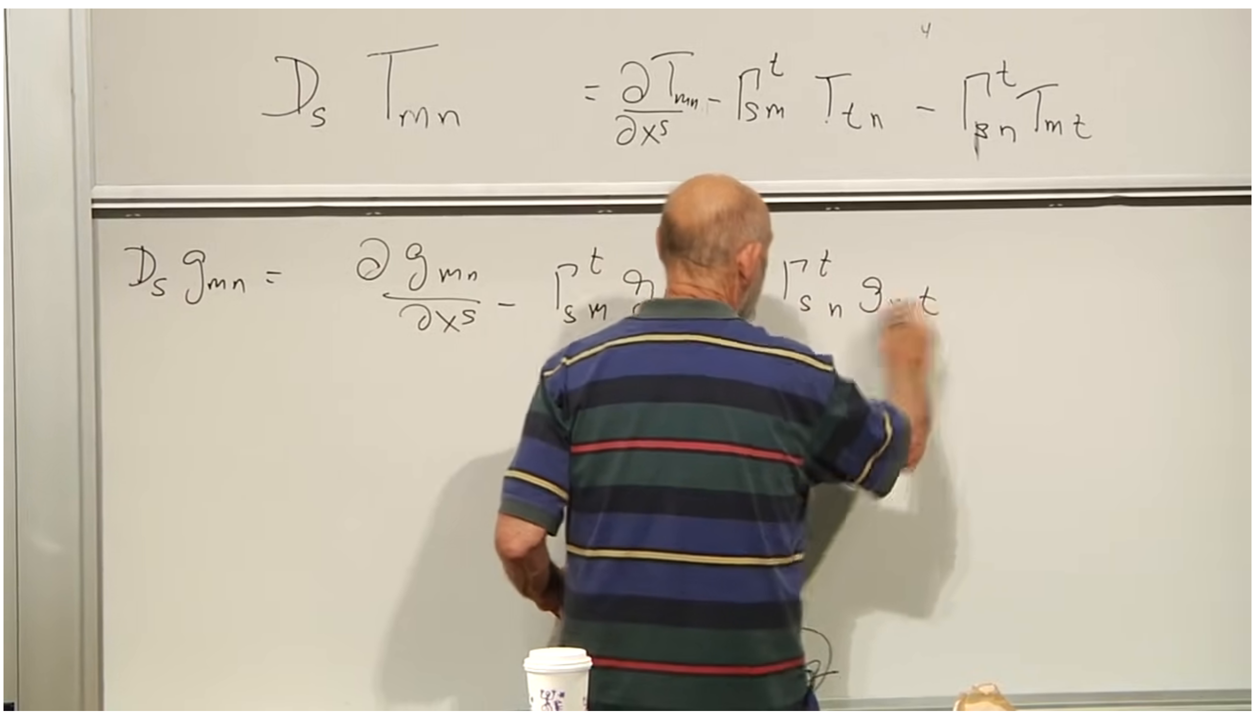

Figure 1.16: The metric-compatibility equation begins the derivation of the connection from the metric.

Lecture frame: 01:06:20.

Expanding (1.14) gives

$$0 = \partial_s g_{mn} - \Gamma^t{}_{sm} g_{tn} - \Gamma^t{}_{sn} g_{mt}. \tag{1.15}$$

Write two more copies with cyclically permuted derivative and metric indices:

$$0 = \partial_m g_{sn} - \Gamma^t{}_{ms} g_{tn} - \Gamma^t{}_{mn} g_{st}, \tag{1.16}$$

$$0 = \partial_n g_{sm} - \Gamma^t{}_{ns} g_{tm} - \Gamma^t{}_{nm} g_{st}. \tag{1.17}$$

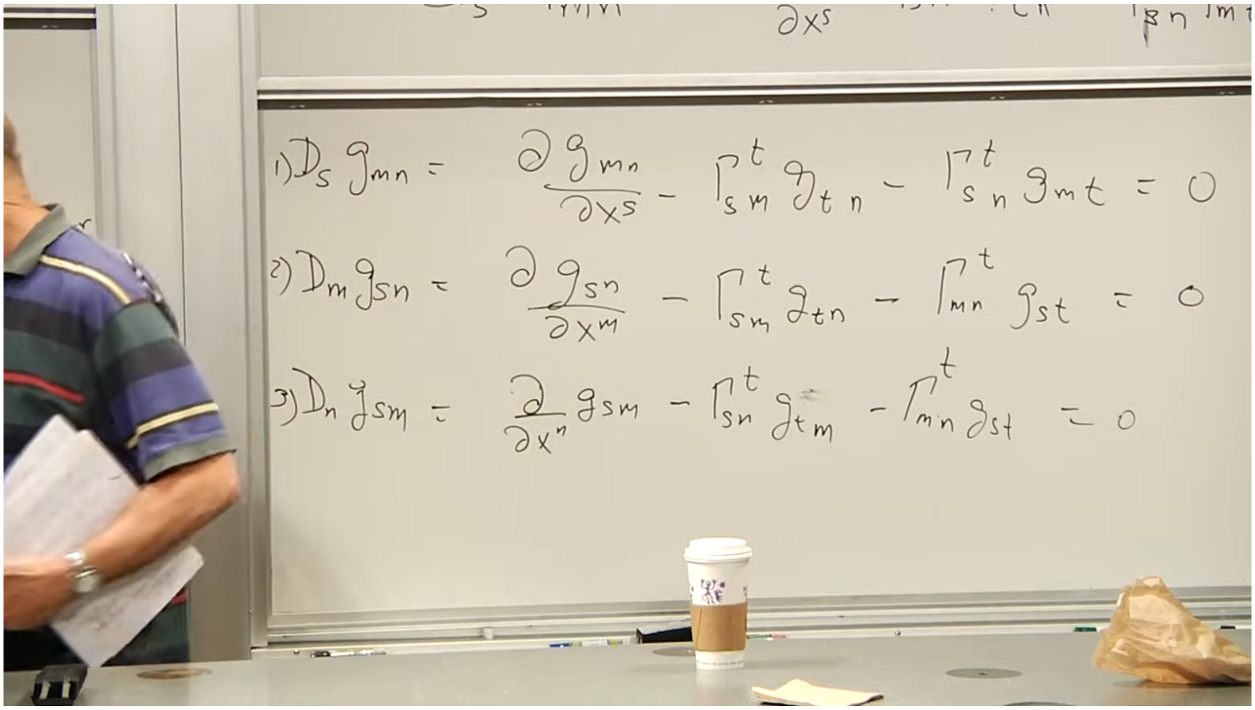

Figure 1.17: The three permuted metric-compatibility equations.

Lecture frame: 01:10:20.

Add (1.16) and (1.17), then subtract (1.15). Using the torsion-free symmetry (1.12), the unwanted connection terms cancel:

$$\partial_m g_{sn} + \partial_n g_{sm} - \partial_s g_{mn} = 2\Gamma^t{}_{mn} g_{st}. \tag{1.18}$$

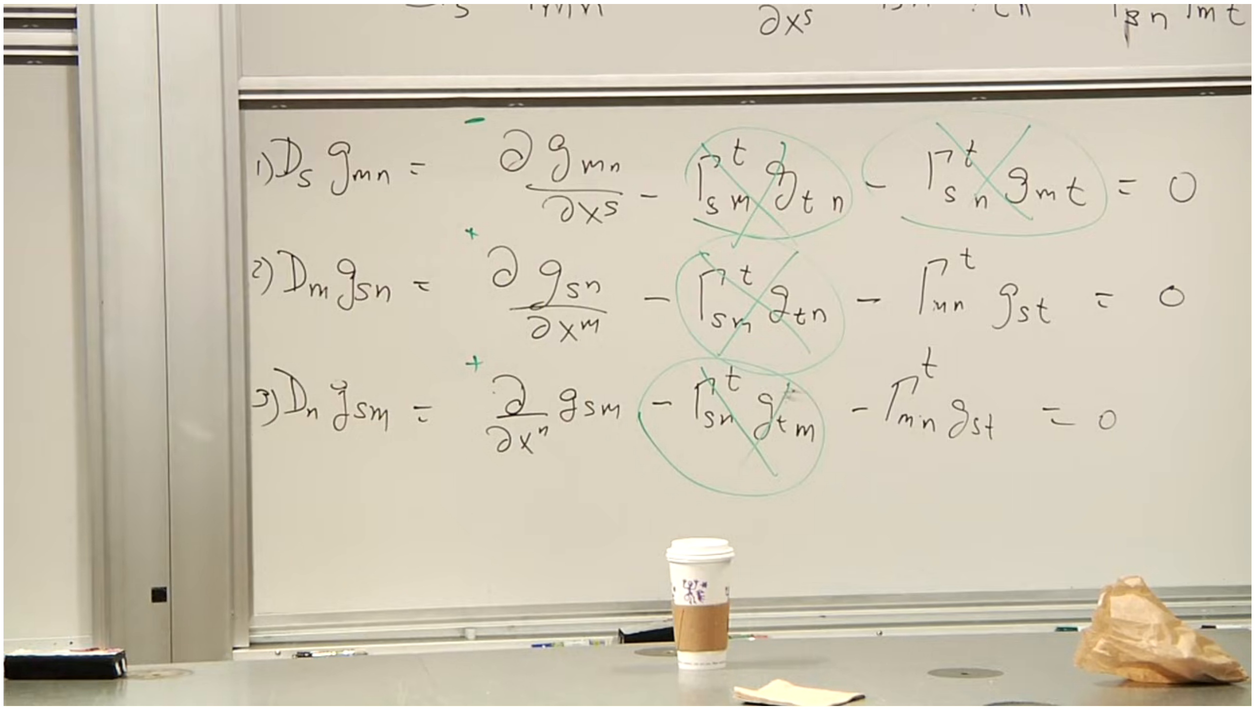

Figure 1.18: Adding two compatibility equations and subtracting the third cancels the unwanted Christoffel terms.

Lecture frame: 01:12:10.

Multiplication by g^{rs} isolates the connection:

$$\boxed{\Gamma^r{}_{mn} = \frac{1}{2} g^{rs} \left(\partial_m g_{sn} + \partial_n g_{sm} - \partial_s g_{mn}\right)} \tag{1.19}$$

Thus the Levi-Civita connection is built from the metric and its first derivatives. This is the promised reason that the Christoffel symbols vanish at the center of normal coordinates.

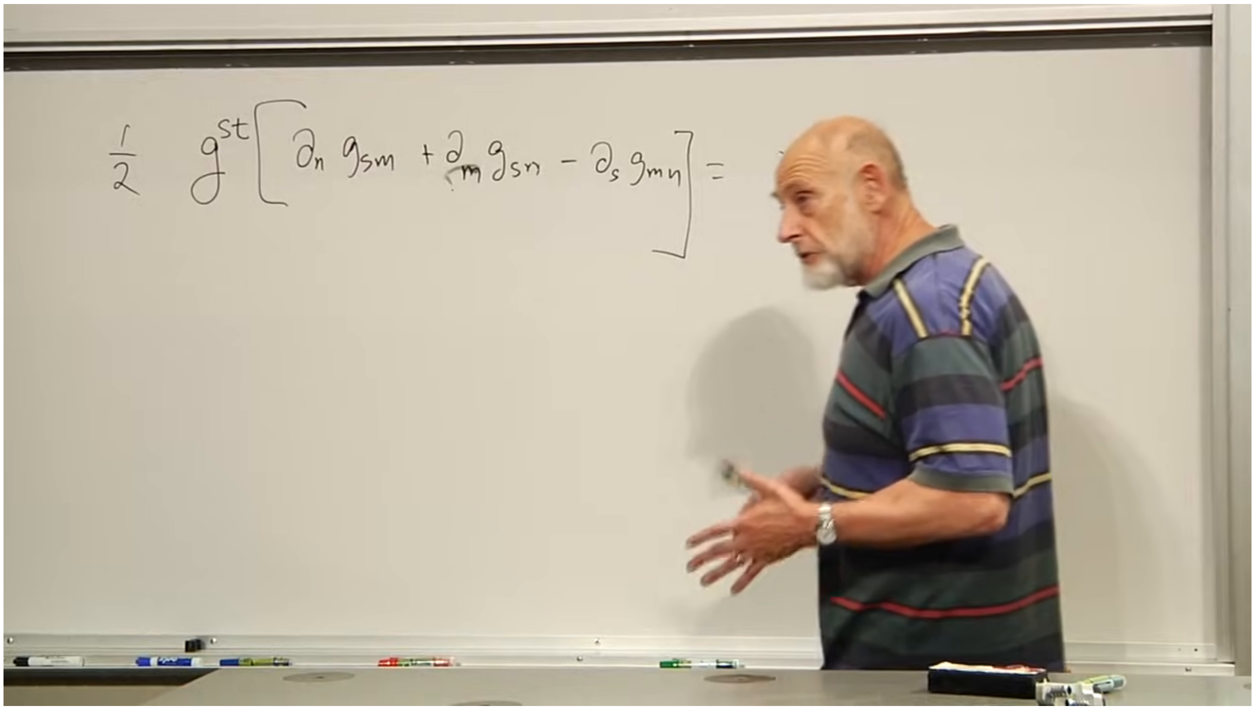

Figure 1.19: The inverse metric isolates the Christoffel symbol from the lowered connection equation.
Lecture frame: 01:16:10.

Question from the audience. Why were all three metric-compatibility equations set equal to zero?

Response. At the selected point in normal coordinates, the ordinary derivatives of the metric vanish and so do the Christoffel symbols. Hence the covariant derivative of the metric is zero there. To establish the statement at any other point, center a normal frame at that point and repeat. Because $\nabla_s g_{mn}$ is a tensor, its vanishing is coordinate independent. [a]

[a]Lecture timestamp: 01:14:27.

Question from the audience. What are the dimensions of the metric and the Christoffel array, and why are there forty independent Christoffel components in four dimensions?

Response. In d dimensions, g_{mn} is a symmetric $d \times d$ matrix. The connection has three indices. For each fixed upper index r, the symmetric lower pair mn has $d(d+1)/2$ choices. Thus a torsion-free connection has $d^2(d+1)/2$ independent components, which is forty when $d = 4$. [a]

[a]Lecture timestamp: 01:16:18.

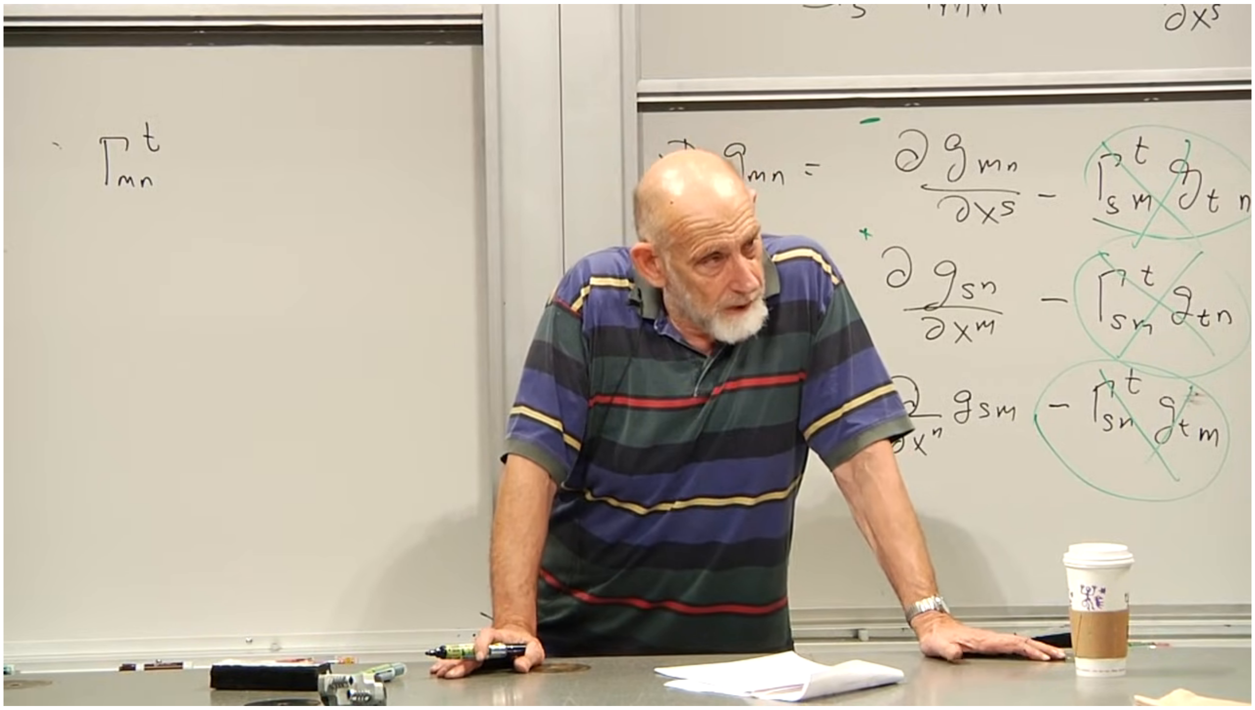

Figure 1.20: The board during the component-count discussion: ten choices for the symmetric lower pair and four choices for the upper index.

Lecture frame: 01:16:30.

Question from the audience. Which metric appears in (1.19)? Must a different normal-coordinate metric be used at every point?

Response. The formula uses the components of the metric in whatever global or local coordinates one has chosen for the calculation. Normal coordinates are only the local device used to derive the rule: move to a normal frame at one point, differentiate there, transform the result back, and then discard the auxiliary frame. Once (1.19) is known, it is evaluated directly in the arbitrary coordinates of interest. [a]

[a]Lecture timestamp: 01:17:21.

Question from the audience. Does either $\nabla_s g_{mn} = 0$ or the Christoffel formula tell us that the space is flat?

Response. No. Metric compatibility holds for every Levi-Civita connection, curved or flat. Christoffel symbols can be nonzero even on a flat plane written in polar coordinates. If all $\Gamma^r{}_{mn}$ vanish throughout a region in one chart, that chart is Cartesian there and the region is flat; but their nonvanishing alone does not diagnose curvature. A tensor built from derivatives and products of Γ is still needed. [a]

[a]Lecture timestamp: 01:19:27.

1.8 A Cone Reveals What the Connection Cannot

Consider a rounded cone. Away from its tip it can be approximated by an ordinary cone made from paper. Cut a wedge from a flat disk and glue the exposed edges together. Nothing in the remaining paper has been stretched; locally, every point away from the join and tip is still flat. The missing wedge is recorded by a deficit angle Δ.

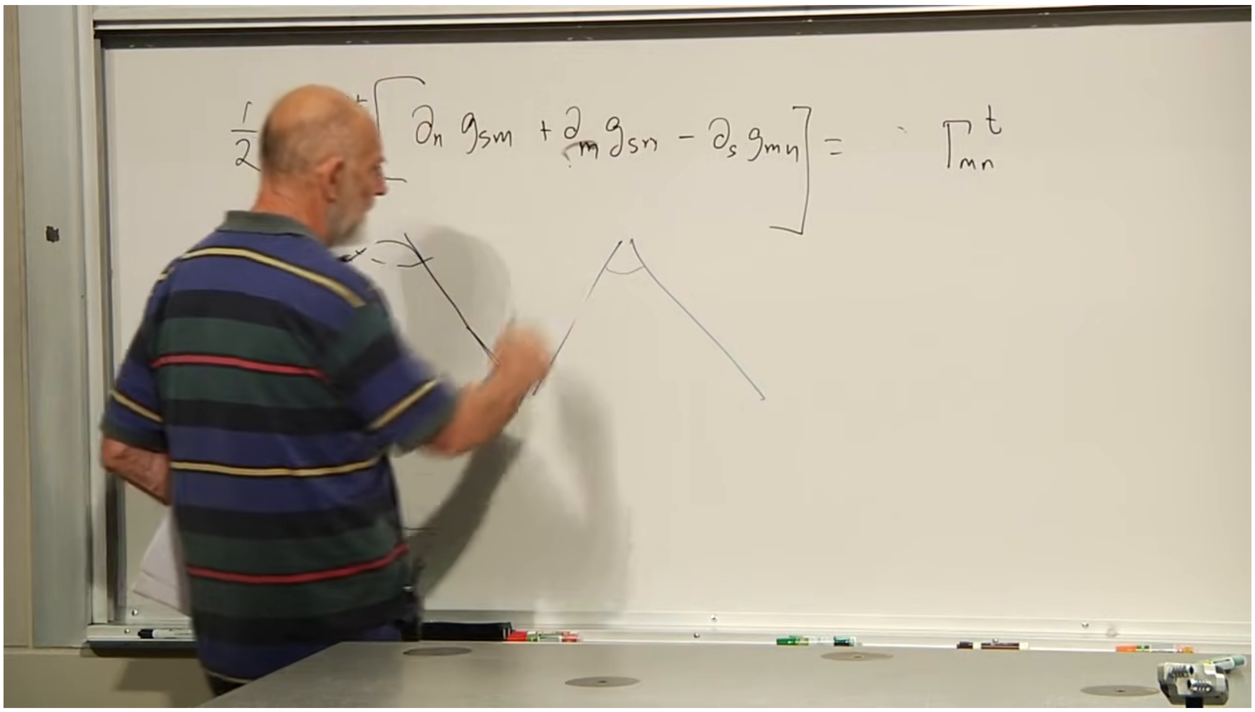

Figure 1.21: The cone introduced as a geometry that is locally flat away from its tip but has nontrivial curvature concentrated at the tip.

Lecture frame: 01:23:00.

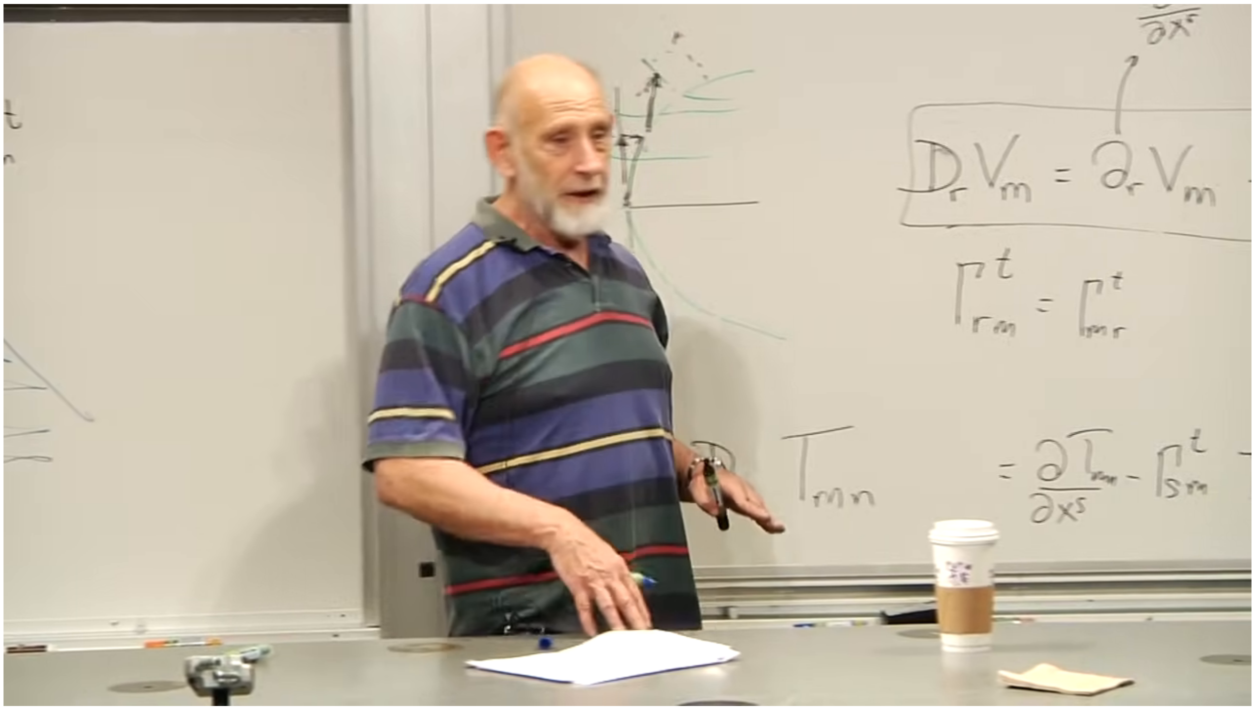

Figure 1.22: Cutting a wedge from a disk and identifying its edges produces the conical deficit angle.

Lecture frame: 01:24:20.

The paper construction makes the distinction tangible. On the cut sheet, transport a vector while keeping it parallel in the ordinary plane. After the wedge edges are identified, a path that circles the tip returns to its starting point with the transported vector rotated relative to its initial direction.

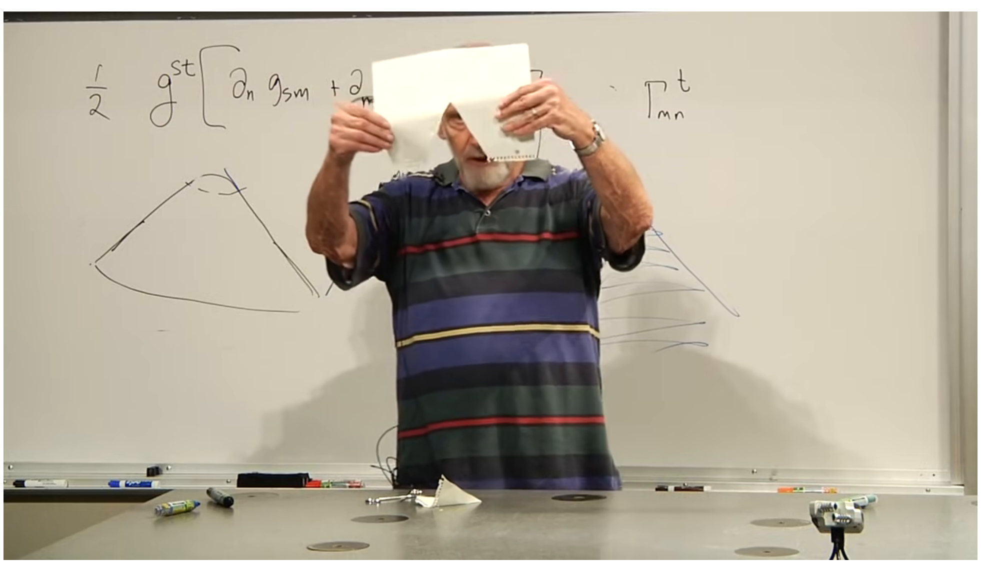

Figure 1.23: The paper-cone demonstration: a flat sheet with a wedge removed becomes a cone when its cut edges are joined.

Lecture frame: 01:25:10.

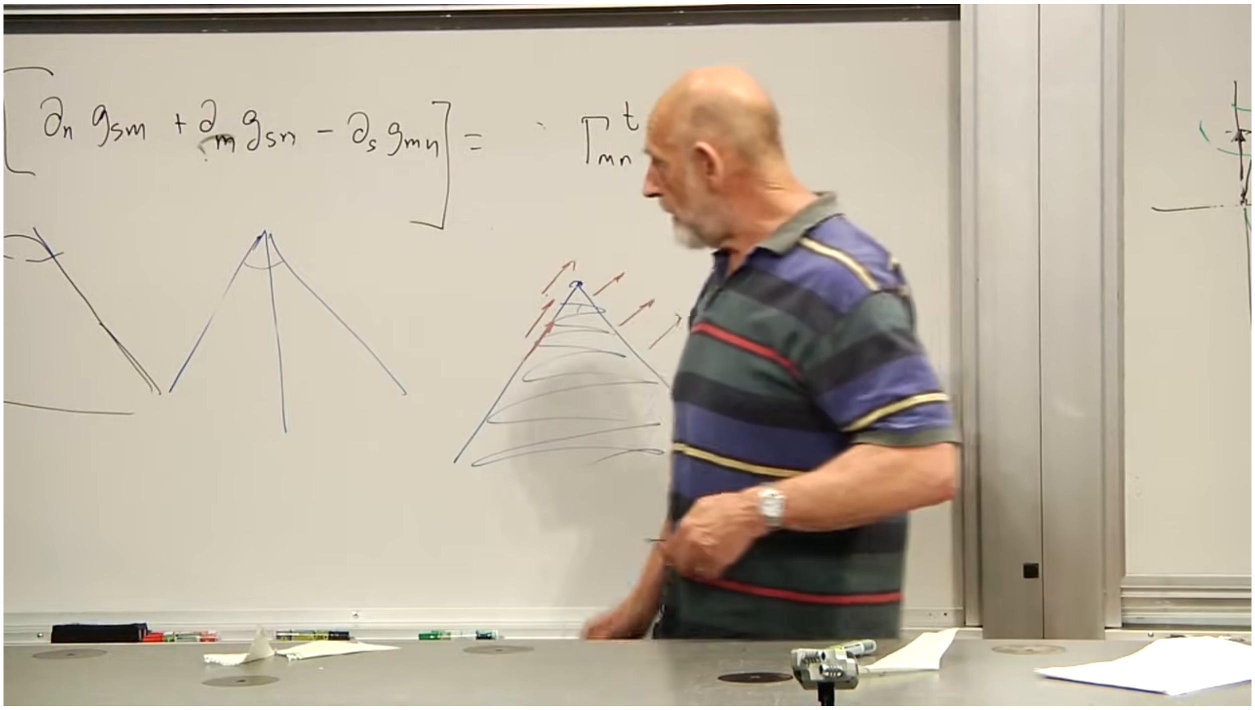

Figure 1.24: Vectors drawn on the cut wedge before the edges are identified.

Lecture frame: 01:27:40.

This rotation is holonomy. It is not caused by a poor choice of coordinates along one segment. Each small segment can be treated in a locally flat frame. The effect appears only after those local comparisons are composed around a closed loop. For the ideal cone the net rotation equals the deficit angle, up to orientation.

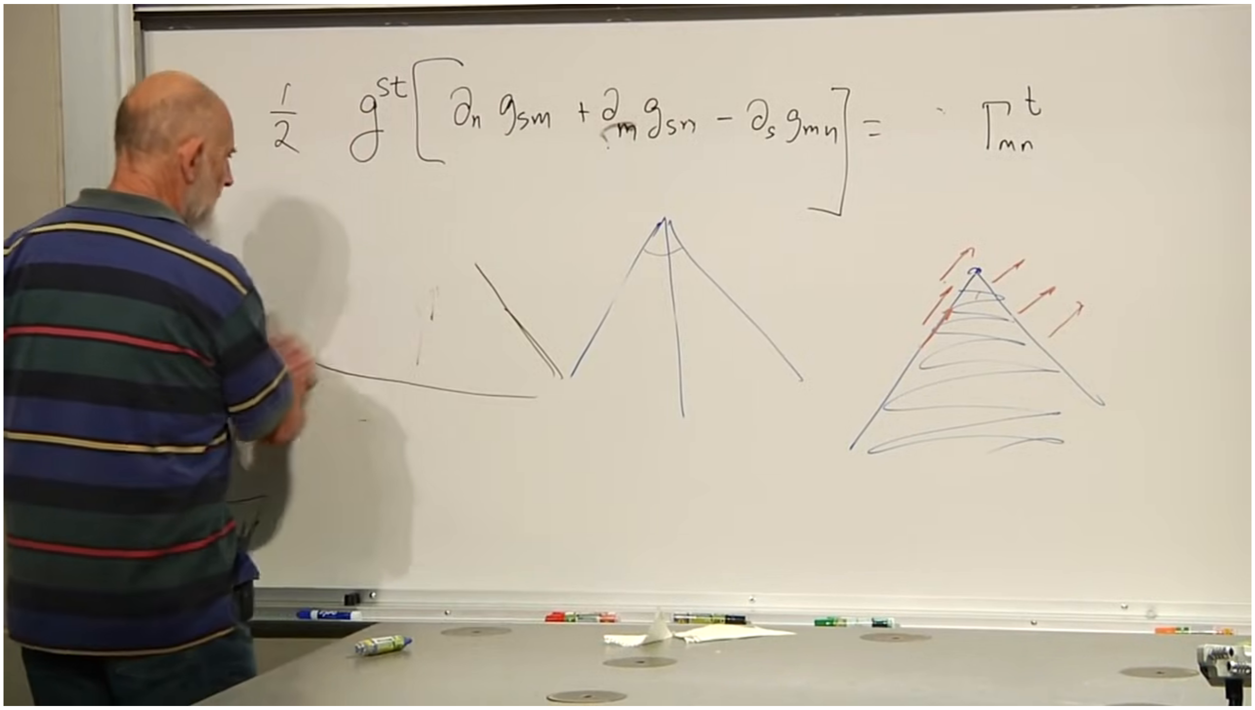

Figure 1.25: Parallel transport around the conical tip returns a vector with a rotated direction: the holonomy records the enclosed curvature.

Lecture frame: 01:29:10.

1.9 Two Routes Around an Infinitesimal Loop

Replace the finite loop by a tiny coordinate rectangle. Starting at one corner, transport first in the r direction and then in the s direction. Compare that result with transport first in s and then in r. In flat Cartesian coordinates the two routes agree. In a curved geometry they generally do not.

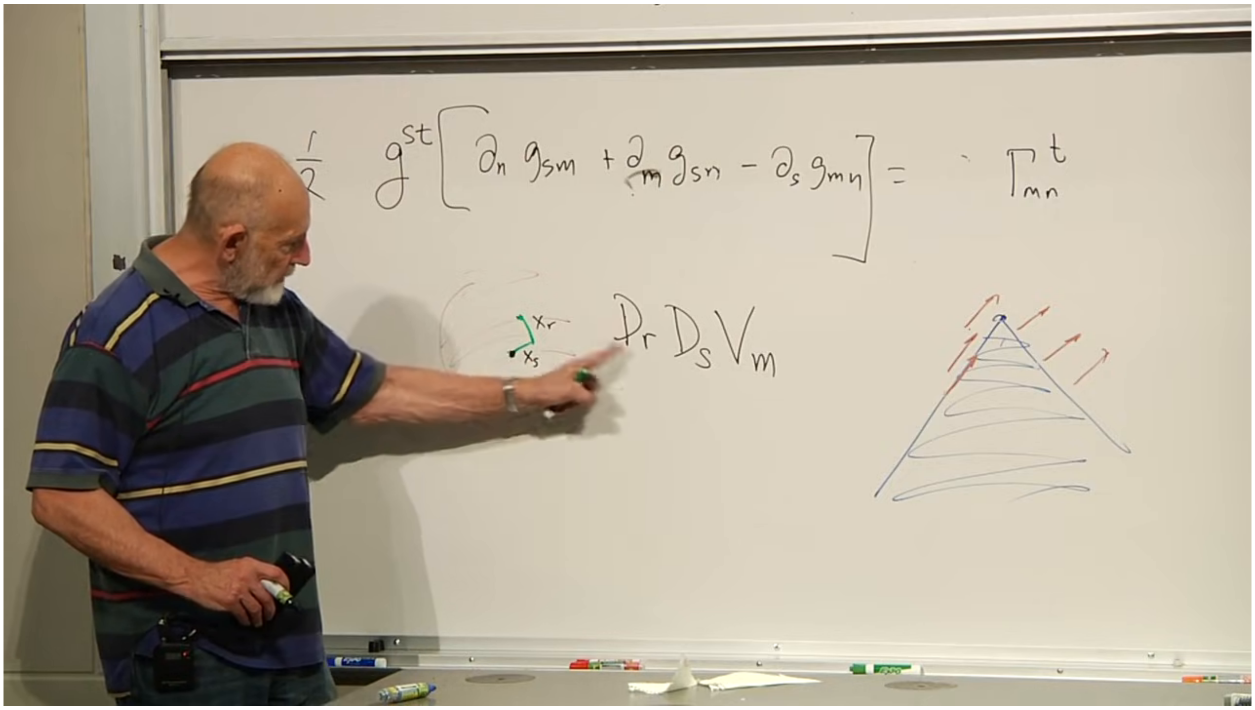

Figure 1.26: Two orderings of infinitesimal transport around a coordinate rectangle.

Lecture frame: 01:31:40.

The algebraic measure is the commutator of covariant derivatives:

$$(\nabla_s \nabla_r - \nabla_r \nabla_s)\, V_n. \tag{1.20}$$

Ordinary mixed partial derivatives commute on smooth functions. The failure here comes from differentiating the connection terms as well as the vector components.

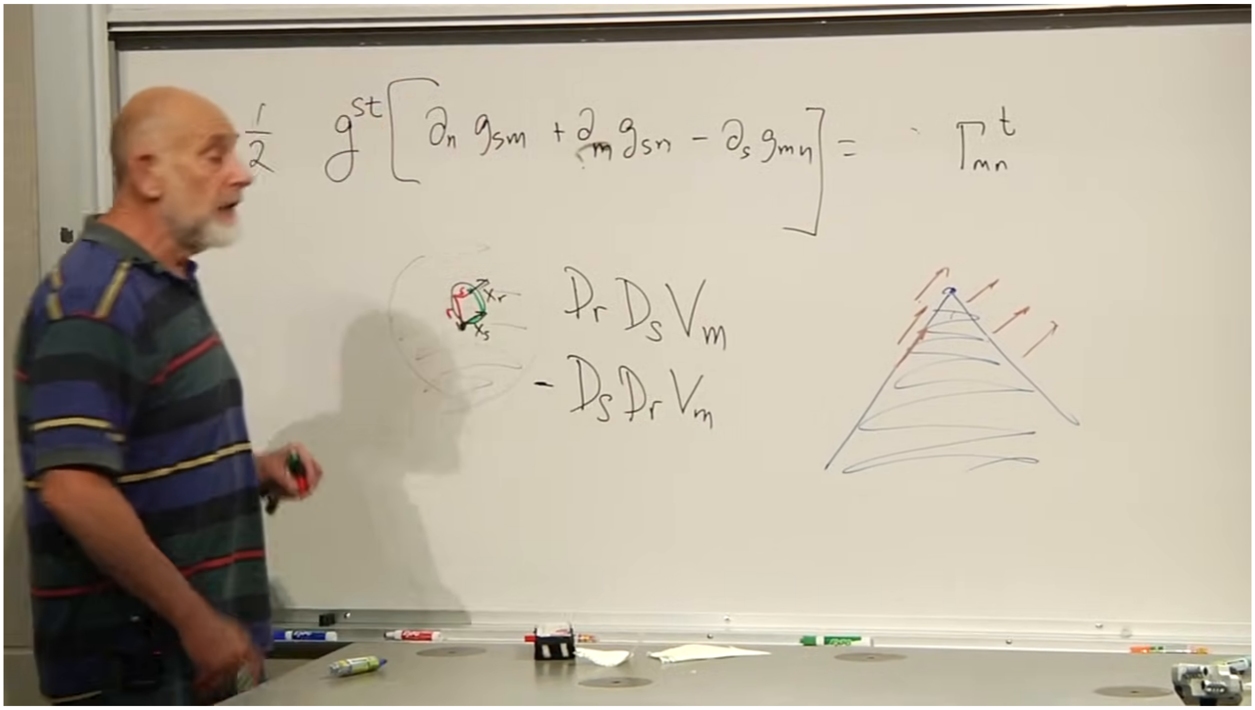

Figure 1.27: The infinitesimal loop represented by the difference between the two orders of covariant differentiation.

Lecture frame: 01:33:20.

To compute it, first regard

$$T_{rn} = \nabla_r V_n$$

as a rank-two covariant tensor. Then use (1.13) to differentiate T_{rn} in the s direction. Repeat with r and s interchanged and subtract. The operation is lengthy but mechanical: expand, apply the product rule, and cancel the terms containing second partial derivatives of V_n.

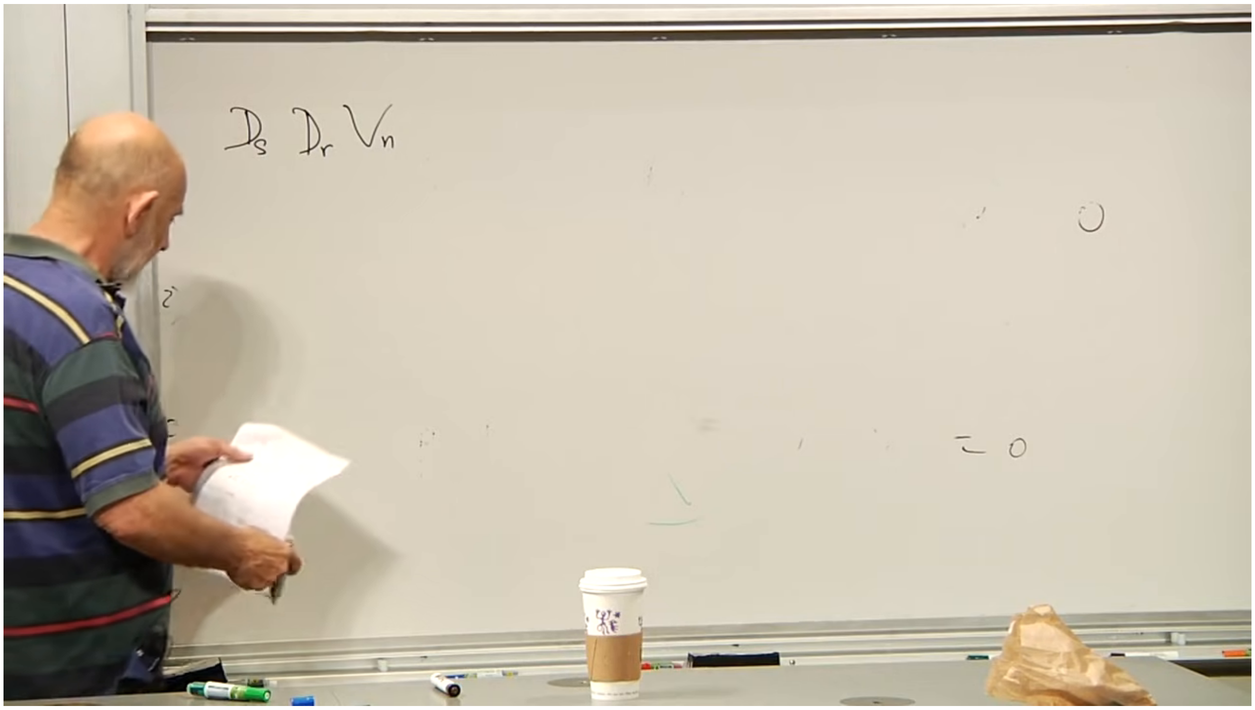

Figure 1.28: Beginning the double covariant derivative by treating $\nabla_r V_n$ as a rank-two tensor.
Lecture frame: 01:35:40.

Using the convention adopted on the board, the result is

$$(\nabla_s \nabla_r - \nabla_r \nabla_s)\, V_n = R_{srn}{}^{t} V_t. \tag{1.21}$$

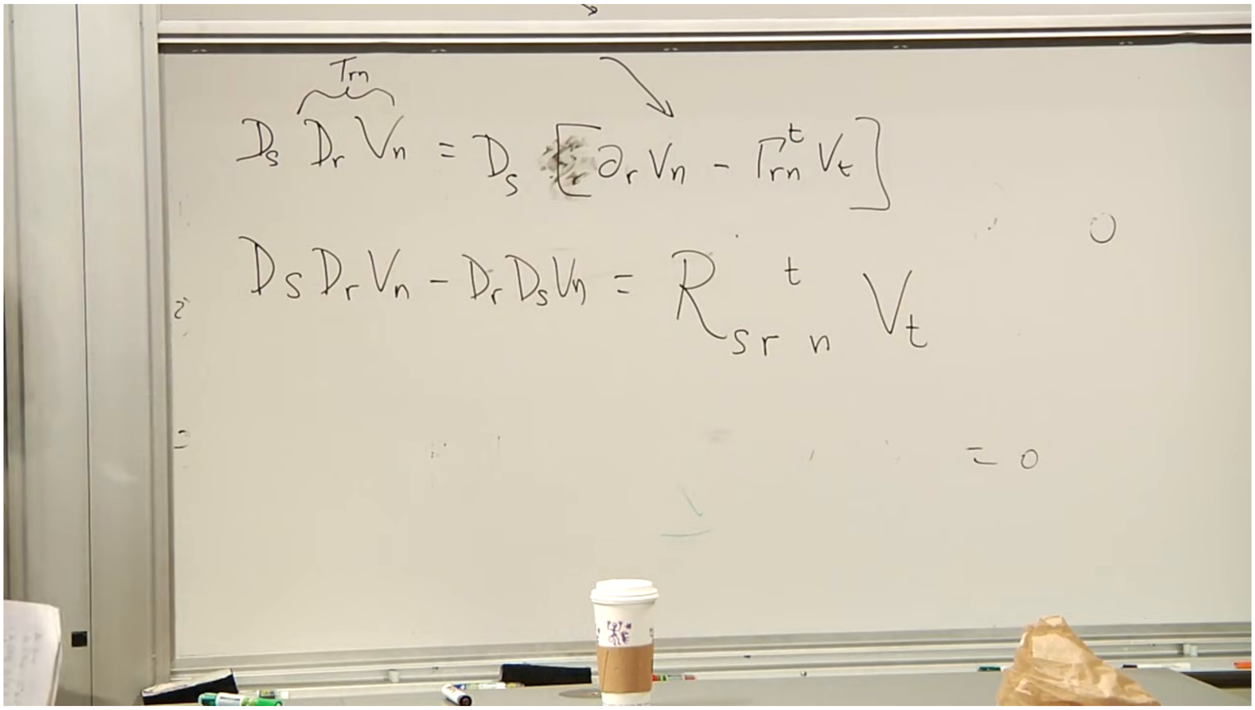

Figure 1.29: The commutator of covariant derivatives defines the action of the curvature tensor on a covector.

Lecture frame: 01:38:30.

For this ordering of the commutator,

$$R_{srn}{}^t = \partial_r \Gamma^t{}_{sn} - \partial_s \Gamma^t{}_{rn} + \Gamma^p{}_{sn}\Gamma^t{}_{pr} - \Gamma^p{}_{rn}\Gamma^t{}_{ps}. \tag{1.22}$$

Other texts may reverse the sign or arrange the indices differently; the convention is fixed by writing both (1.21) and (1.22).

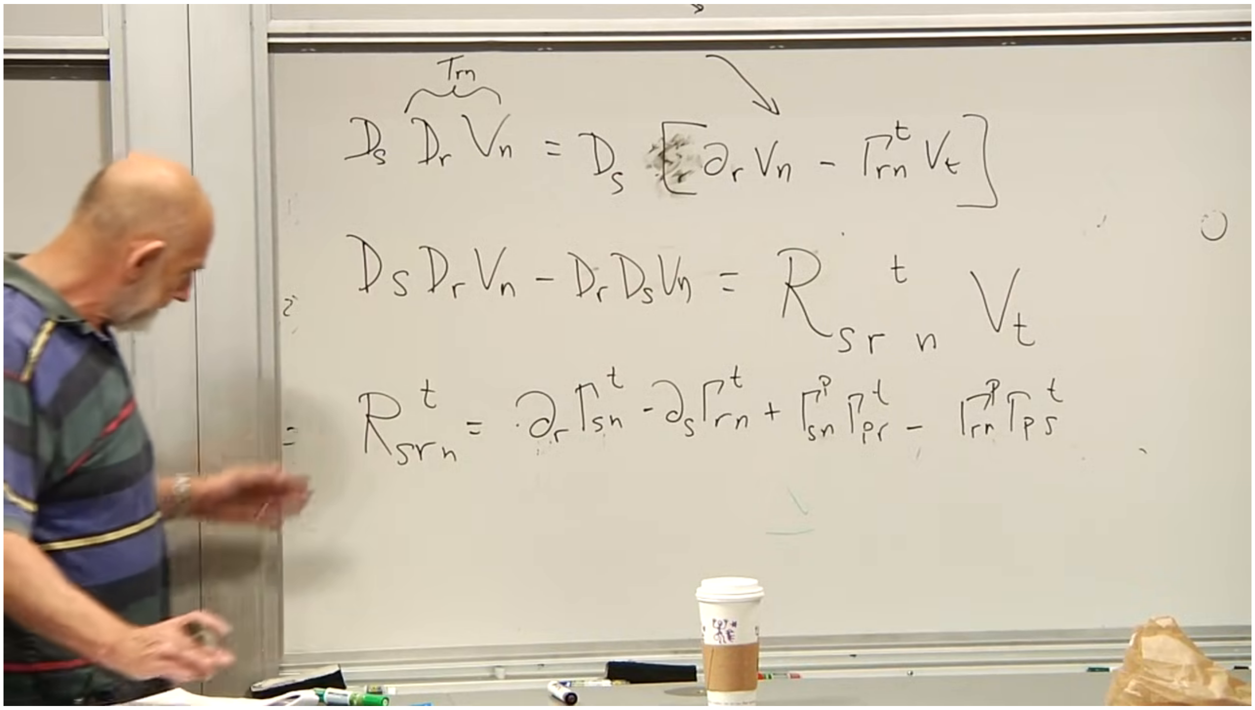

Figure 1.30: The Riemann tensor: two derivatives of the connection and two quadratic connection terms.

Lecture frame: 01:39:40.

Equation (1.19) shows why curvature first appears at second order in the metric. Each Γ contains a first derivative of g. The derivative terms in (1.22) therefore contain second derivatives of g, while the quadratic terms contain products of first derivatives. In normal coordinates at P, the connection itself vanishes but its derivatives generally do not, so the curvature survives:

$$\Gamma^t{}_{mn}(P) = 0, \qquad R_{srn}{}^t(P) = \partial_r \Gamma^t{}_{sn}(P) - \partial_s \Gamma^t{}_{rn}(P).$$

It is important to keep the derivative order straight: the Riemann tensor uses first derivatives of Γ, which are second order in the metric.

This is the desired diagnostic. If every component of the Riemann tensor vanishes throughout a region, that region is locally flat. If any component is nonzero at a point, no change of coordinates can remove the curvature there. A computer algebra system can test these components from an analytic metric, but the concept is simply the failure of parallel transport to close around an infinitesimal loop.

1.10 Curvature as Tidal Gravity

The geometric diagnostic has a direct physical meaning. Imagine a flexible object made from linked triangles. In a flat region it can be moved without stress. Push it into a region where the intrinsic metric curves, and different parts of the linkage cannot all retain their preferred separations and angles. The object must stretch, squeeze, or shear.

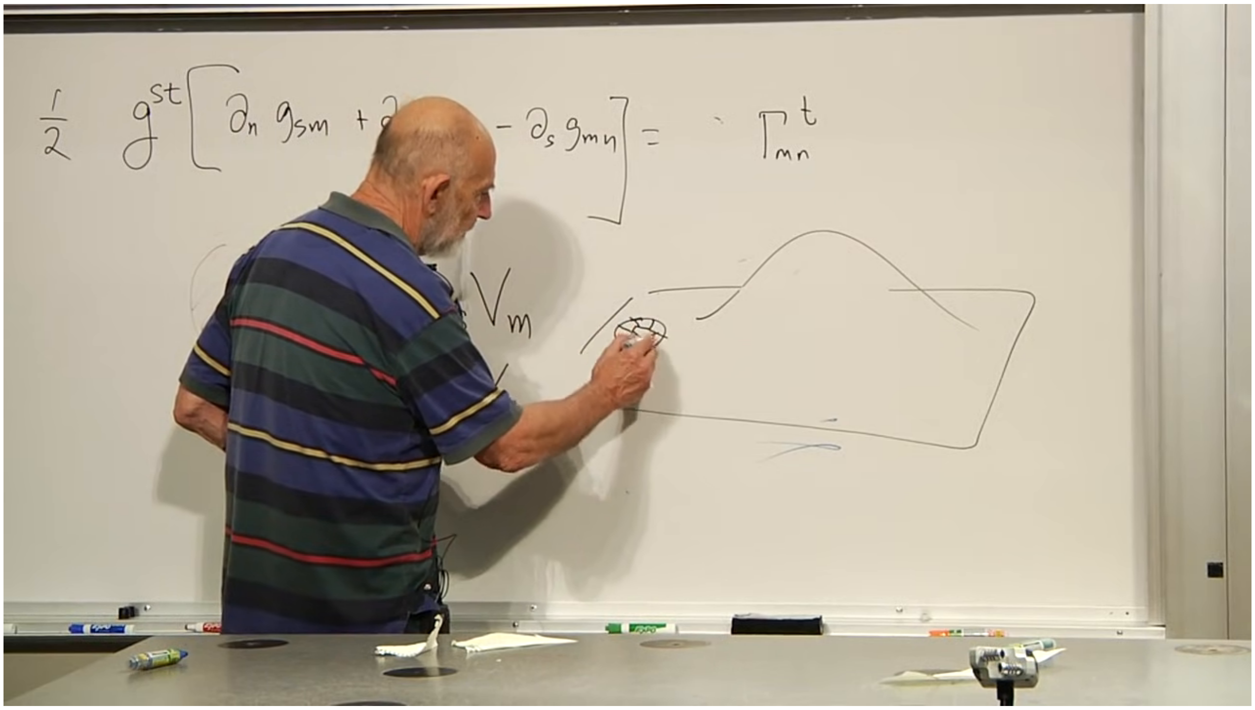

Figure 1.31: A linked object moved from a flat region into a curved bulge must deform to follow the local metric.

Lecture frame: 01:44:10.

That deformation is the geometric counterpart of tidal force. In a gravitational field, neighboring freely falling particles can accelerate relative to one another: one direction may be stretched while another is squeezed. The Riemann tensor controls this relative acceleration. In the language introduced later as geodesic deviation, for tangent u^μ and separation ξ^μ,

$$\frac{D^2\xi^\mu}{D\tau^2} = -R^\mu{}_{\nu\rho\sigma} u^\nu \xi^\rho u^\sigma, \tag{1.23}$$

with the overall sign following the chosen curvature convention. This equation is not derived here, but it makes precise the lecture's statement that curvature and tidal stress are literally the same local physics.

An ideal perfectly uniform gravitational field has no tidal variation and can be removed locally by passing to a uniformly accelerated frame. Real fields produced by isolated masses are not perfectly uniform. Earth's field points approximately radially and changes with position, so neighboring free-fall trajectories converge or separate.

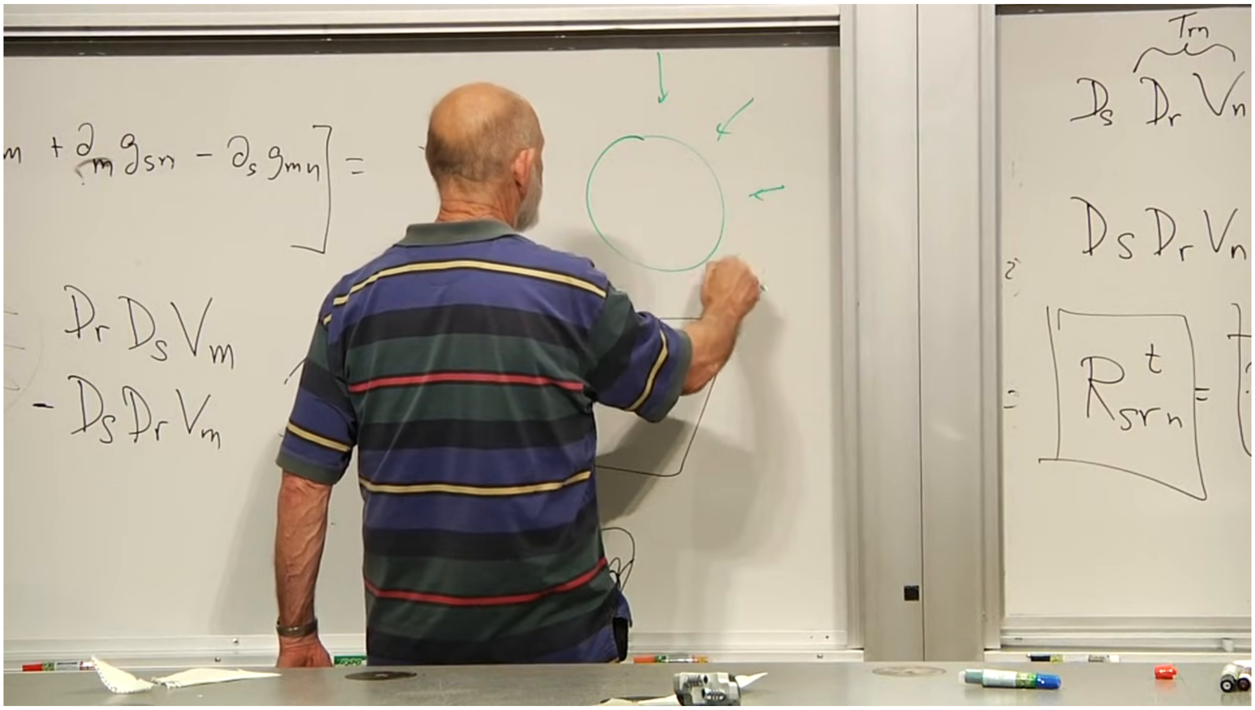

Figure 1.32: A radial gravitational field varies across an extended body, producing the relative acceleration identified as tidal gravity.

Lecture frame: 01:48:40.

Question from the audience. What distinguishes a genuine gravitational field from the apparent field seen in an accelerated frame?

Response. A uniformly accelerated frame can imitate a uniform gravitational field for local experiments, but it does not produce tidal relative acceleration. A field generated by real matter varies from place to place and therefore produces tidal forces. Those forces are invariant evidence that the effect is not merely a coordinate transformation. A perfectly uniform real field is an idealization; near a sufficiently small patch of a large body's surface, the actual field can only be approximately uniform. [a]

[a]Lecture timestamp: 01:46:54.

Question from the audience. Why does the size of the falling object matter for tidal effects?

Response. Curvature compares neighboring locations. An object much smaller than the length scale over which the field changes samples almost the same acceleration at all of its parts and suffers little deformation. An extended object samples a larger difference. A freely falling bacterium near Earth feels negligible tidal stress, whereas the lecture's imaginary two-thousand-mile person would be severely distorted. The curvature has a local value, but the accumulated deformation scales with the separation being compared. [a]

[a]Lecture timestamp: 01:49:01.

Question from the audience. If flatness requires checking the curvature at every point, is the test still an infinite problem?

Response. For a metric given analytically, the derivatives and curvature components are analytic expressions. One calculates them once as functions of position and can often see whether they vanish identically. If the metric is supplied only as a numerical table, then one must estimate and inspect curvature throughout the sampled region, which can indeed be expensive. Even then, this is far better defined than searching every possible coordinate system at every point. [a]

[a] Lecture timestamp: 01:50:37.

1.11 What Has Been Established

The argument can now be read as one continuous chain:

1. At any chosen point, coordinates can make $g_{mn} = \delta_{mn}$ and $\partial_r g_{mn} = 0$.
2. Ordinary derivatives of tensor components fail to transform as tensors because coordinate bases vary from point to point.
3. The connection corrects that basis variation and defines covariant differentiation.
4. Metric compatibility and zero torsion determine the connection uniquely from the metric.
5. Two covariant derivatives fail to commute by the Riemann tensor. That failure measures holonomy, second-order metric variation, and tidal gravity.

The promised curvature diagnostic has therefore been constructed without choosing a privileged global coordinate system. The next step is to change from positive-definite Riemannian geometry to Lorentzian spacetime, apply these tools to the Schwarzschild geometry of a star or black hole, and return to the geodesics that were deferred at the end of this lecture.